

Schooling–Labour Trade-off among Undocumented Immigrants in the US: Evidence from DACA and In-State Tuition Policy

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8th May 2026

Abstract

Undocumented immigrants in the United States face restricted access to formal employment and substantial barriers to higher education, distorting the usual decision between these outcomes. Two policies relaxing these restrictions, “Deferred Action for Childhood Arrivals” (DACA), granting work authorisation, and state-level policy offering in-state resident tuition to undocumented immigrants, provide a meaningful opportunity to study their allocation between school and work. DACA initially increases labour force participation among likely eligible immigrants with little evidence of a contemporaneous rise in college enrolment, while the later uncertainty period attenuates these labour-market gains and decreases college enrolment. Offering in-state resident tuition increases college enrolment, but we find heterogeneous effects by DACA regime. Before DACA, it raises enrolment with only marginal statistical significance and without significant labour-force effects. Once DACA is implemented, college enrolment effects strengthen and we find that reducing schooling costs decreases labour force participation. These results jointly suggest that undocumented immigrants respond along both the return-to-schooling margin and the cost-of-schooling margin, and that work authorisation produces a trade-off between schooling and working.

*I would like to offer special thanks to Dr. Dunli Li for her ongoing support and supervision throughout this project, as well as for comments from Matthew Nibloe, Anton Reinicke, Professor Ian Preston, and Professor Aureo de Paula.

Contents

1	Introduction	3
2	Literature Review	5
2.1	Undocumented Immigrants	6
2.2	Deferred Action for Childhood Arrivals	7
2.3	In-State Resident Tuition	8
3	Theoretical Framework	10
3.1	Schooling Choice Under Uncertain Formal-Sector Entry	10
3.2	Comparative Statics	12
3.3	Risk Aversion and the Uncertainty Period	13
4	Data	15
5	Methodology	17
5.1	DACA: Methodology	17
5.2	In-State Resident Tuition: Methodology	21
6	Results	23
6.1	DACA: Results	23
6.2	In-State Resident Tuition: Results	26
7	Discussion	30
7.1	Interpretation	30
7.2	Policy Implications	31
7.3	Limitations	33
8	Conclusion	34
9	Appendix	42

1 Introduction

Undocumented immigrants in the United States face barriers to both higher education and labour market participation. In the labour market, the absence of legal work authorisation restricts access to formal employment, depressing economic opportunities (U.S. Citizenship and Immigration Services 2025b). In higher education, undocumented students often face substantially higher tuition charges at public colleges and universities, are excluded from federal financial aid, and frequently come from low-income households under pressure to contribute financially (Federal Student Aid 2026; Broder 2025). These constraints may reduce schooling or work in isolation, as well as distort the allocation between them. For undocumented immigrants, the relevant choice is often not the standard one between schooling and work under open opportunity; undocumented immigrants decide under limited access to lawful employment, high direct schooling costs, and short planning horizons.

These facts may lead one to expect that undocumented immigrants should respond strongly to policies that change either the returns to work or the cost of schooling. Deferred Action for Childhood Arrivals (DACA) relaxed barriers to formal employment by granting temporary work authorisation and protection from deportation. In-state resident tuition policies lowered the direct price of college by allowing undocumented students to pay resident rather than out-of-state tuition. A large literature studies each margin, but the evidence is mixed. On the labour-market side, DACA is associated with higher labour force participation and employment, yet its effects on college enrolment have no such consensus. Some studies find little evidence of changes in school attendance, others find declines in enrolment as work becomes more attractive, and others find heterogeneous responses for four-year university and community college students (Pope 2016; Amuedo-Dorantes and Antman 2017; Hsin and Ortega 2018; Hamilton, Patler, and Savinar 2020). On the schooling-cost side, several studies find that in-state tuition raises college enrolment among likely undocumented youth, while others find weaker or null effects (Kaushal 2008; Flores 2010a; Chin and Juhn 2010; Amuedo-Dorantes and Sparber 2014).

Yet, this literature has largely treated these policies separately. That is a consequential omission. A work-authorisation policy may only induce enrolment into higher education if the recipients can afford the investment. A tuition policy changes schooling decisions differently depending on whether their skills can be applied in formal labour markets. In other words, the effect of lowering the cost of college should depend on the return to schooling, and the effect of raising the return to work should depend on the cost of acquiring more education. The interaction between these two margins remains underdeveloped in the literature, even though it is central to understanding how undocumented immigrants allocate themselves

between school and work.

In this paper, we ask whether undocumented immigrants respond to incentives operating through labour market returns and the cost of schooling, and what those responses reveal about the mechanisms governing their higher education and labour supply decisions. We study this question by combining two policy shocks that relax different constraints. DACA expands access to formal employment and raises the payoff to labour market participation. In-state resident tuition lowers the direct cost of college. We then assess how these effects vary under a period of policy uncertainty for DACA created by the Trump administration’s attempt to rescind DACA following the 2016 election. Using repeated cross-sections from the American Community Survey and exploiting variation across states and over time, we examine how these policies shape college enrolment and labour force participation among “likely undocumented” Mexican-born non-citizens, with particular attention to how the effects of tuition policy vary across DACA regimes.

We find a clear schooling–labour trade-off once we assess the policies together. DACA in isolation initially increases labour force participation by about 7 percentage points, with little evidence of a contemporaneous increase in college enrolment. In the later uncertainty period post-2016, the labour-market effect weakens to roughly 4.5 percentage points relative to before DACA’s implementation while college enrolment falls by about 6.5 percentage points. In-state resident tuition, by contrast, increases college enrolment throughout, but its effects depend on the surrounding DACA regime. Before DACA, lower tuition raises enrolment modestly, by about 3.6 percentage points, with no clear labour-force response. Once DACA is in place, the college effect strengthens to about 6.8 percentage points and labour force participation now falls by about 7.8 percentage points. In the uncertainty period, the college effect remains positive but smaller, around 4.3 percentage points, while the negative labour-force effect persists. These results suggest that undocumented immigrants respond along both the return-to-schooling margin and the cost-of-schooling margin, and that work authorisation creates a meaningful trade-off between schooling and work.

These findings make three contributions. First, the paper helps reconcile the mixed evidence on DACA and college enrolment by showing that DACA’s educational effects depend on the broader schooling-cost environment. DACA changes the expected payoff to labour force participation by making formal work accessible and, through that channel, the return to schooling; in-state resident tuition lowers the direct cost of acquiring that schooling. Studying the two policies together therefore reveals a schooling–labour trade-off that is difficult to see when each policy is considered in isolation. Second, the paper extends the DACA literature by studying the college enrolment and labour force participation outcomes

and distinguishing the initial implementation period from the later period of legal and political uncertainty. Third, the paper revisits the DACA and in-state resident tuition question using a tighter proxy for undocumented status and modern staggered-adoption methods designed for heterogeneous treatment effects. The central contribution is to show how work authorisation and schooling costs jointly shape the allocation of undocumented immigrants between education and work.

We motivate this study with recent legal and political developments, which have made both DACA and in-state resident tuition especially salient objects of study. DACA was created by executive action, never codified in statute, and has remained legally fragile since the Trump administration’s 2017 rescission attempt, leaving recipients in what Hamilton, Patler, and Savinar (2020) describe as a condition of liminal legality, in which rights remain temporary and contested. In-state resident tuition policies have likewise proven politically unstable, where recent reversals in Texas, Florida, and Oklahoma show that tuition access for undocumented students cannot be treated as fixed institutional background (U.S. Citizenship and Immigration Services 2017; U.S. Citizenship and Immigration Services 2025a; National Immigration Law Center 2025; U.S. Department of Justice 2025; Texas Tribune 2025; Florida Senate 2025; Office of the Oklahoma Attorney General 2025; Oklahoma State Regents for Higher Education 2025). More broadly, these challenges have unfolded amid a stricter enforcement climate during President Trump’s terms (Department of Homeland Security 2025a; Department of Homeland Security 2025b). Moving forward, understanding how these policies adjust the schooling-labour choice for this population will allow us to interpret these recent developments in terms of schooling and labour responses.

The remainder of the paper proceeds as follows. Section 2 reviews the literature on the schooling–labour trade-off facing undocumented immigrants and the existing evidence on DACA and in-state resident tuition. Section 3 provides a theoretical framework for the choice of higher education under uncertain formal employment. Section 4 describes the data and variable construction. Section 5 outlines the empirical strategy. Section 6 presents the main results. Section 7 provides an interpretation of the results, the policy implications of the findings, and a discussion of the limitations. Section 8 concludes.

2 Literature Review

We structure this literature review by theme as follows: (1) Distorted schooling-labour trade-off for undocumented immigrants, (2) existing policy assessment of DACA, and (3) existing policy assessment of in-state resident tuition.

2.1 Undocumented Immigrants

Undocumented immigrants face a labour market that differs fundamentally from that of legal immigrants and natives. Existing work shows that undocumented status is associated with weaker formal employment opportunities and a wage penalty, implying that the work margin is not governed by the same returns as in the standard schooling–labour choice (Rivera-Batiz 1999; Borjas and Cassidy 2019). Relatedly, Borjas (2017) shows that undocumented men exhibit unusually high labour force participation, while undocumented women participate at lower rates, suggesting that undocumented households adjust labour supply in distinctive ways under legal and economic constraints.

Evidence on amnesty and enforcement further underscores that undocumented immigrants face distorted incentives on both the schooling and labour-market margins. For schooling outcomes, the Immigration Reform and Control Act (IRCA) in 1986 provides early evidence of amnesty expanding educational access. Immigrant youth who benefited from IRCA were more likely to enrol in higher education, suggesting that legal status can raise the expected return to further schooling (Cortes 2013). By contrast, enforcement measures move in the opposite direction. Local ICE partnerships reduced Hispanic public-school enrolment, with effects concentrated among elementary-school students (Dee and Murphy 2019), and intensified interior immigration enforcement increased grade retention and school dropout among children with likely unauthorized parents, especially younger children (Amuedo-Dorantes and Lopez 2015). The literature suggests that legal insecurity can disrupt school progression through financial constraints, family separation, residential instability, and fear of deportation.

The labour-market evidence points to a similar pattern. Early work on employer sanctions emphasized that sanctions operate like a tax on the hiring of unauthorized workers, reducing labour demand in the sectors most exposed to enforcement (Hill and Pearce 1990). Empirically, sanctions lowered wages while legalization raised them in metropolitan manufacturing labour markets (Cobb-Clark, Shiells, and Lowell 1995). More recent enforcement evidence points in the opposite direction, as Secure Communities reduced the employment of likely undocumented immigrants, not only through removals but also through “chilling effects” on those who remained in the United States (East et al. 2022). Legalization has been shown to produce the opposite effect. IRCA raised wages and reduced the wage penalty associated with unauthorized status (Kossoudji and Cobb-Clark 2002), and had heterogeneous effects by gender, with falling employment but faster wage growth consistent with improved job mobility or higher reservation wages after legalization (Amuedo-Dorantes, Bansak, and Raphael 2007). NACARA, a policy providing legal permanent residence to certain national-

ities, increased wages and weekly earnings despite only modest employment effects (Kaushal 2006).

The literature shows that undocumented status distorts both schooling and labour-market decisions, but the two margins are usually studied in isolation through broad amnesty such as IRCA. Our paper contributes by exploiting variation from separate policies with different timings to identify how undocumented immigrants reallocate between college enrolment and labour-force participation when employment restrictions and schooling costs change independently. This allows us to move beyond asking whether legal status improves schooling or labour outcomes on average, and instead to estimate the schooling–labour trade-off directly.

2.2 Deferred Action for Childhood Arrivals

DACA provides temporary protection from deportation and renewable two year access to formal labour markets. We review the literature on its effects on labour force participation and educational investment, as well as how these effects change with the policy environment surrounding DACA.

Early evidence on DACA points to large and immediate positive effects on labour market participation. Using American Community Survey (ACS) data, Pope (2016) finds that DACA increased labour-force participation and using Current Population Survey (CPS) data, Amuedo-Dorantes and Antman (2017) provide supporting evidence, reporting improved employment outcomes. Related work finds strong educational responses at the high-school margin, as Kuka, Shenhav, and Shih (2020) and Ballis (2025) find that DACA increased high-school attendance and graduation. DACA also generated broader socio-economic gains, reducing poverty, expanding health insurance coverage, improving mental and intergenerational health, and reducing property crime (Amuedo-Dorantes and Antman 2016; Venkataramani et al. 2017; Giuntella and Lonsky 2020; Hainmueller et al. 2017; Hamilton, Langer, and Patler 2021; Gunadi 2020). By contrast, the initial evidence on college enrolment is more ambiguous. Pope (2016) finds little effect on school attendance, whereas Amuedo-Dorantes and Antman (2017) find that DACA reduced enrolment among higher-educated likely eligibles, interpreted as the effect of an increase in the opportunity cost of college. Hsin and Ortega (2018) similarly show that temporary work authorization often shifted students toward work over higher education investment for those enrolled at four-year colleges, with weaker effects for those at community college. Using California Health Interview Study (CHIS) data, Hamilton, Patler, and Savinar (2020) find insignificant effects on college enrolment, and reconcile the mixed findings in the literature using survey

and interview methods to clarify mechanisms, finding that DACA enabled college for some recipients but discouraged it for others.

Evidence on the uncertainty period is still limited, especially for educational outcomes directly. Amuedo-Dorantes and Wang (2025) show that the legal and political uncertainty surrounding DACA after 2017 eroded some of the program’s early labour-market gains, with employment and income benefits weakening under the post-2016 threat to terminate the policy, particularly in less supportive local environments. At the same time, they argue that gains in education were more durable, suggesting that some of DACA’s effects operated through longer-lasting changes in behaviour and expectations. Related work points in the same direction. Amuedo-Dorantes and Wang (2024) find that, once DACA was threatened, eligible immigrants became more likely to marry U.S. citizens, consistent with recipients responding to heightened uncertainty by seeking more secure legal status through family formation. Wang, Winters, and Yuan (2022) show that DACA increased homeownership among eligible household heads, which is suggestive of stronger incentives to undertake long-term investments when deportation risk falls and expected duration in the United States rises. Likewise, studies documenting increased health insurance coverage after DACA, such as Giuntella and Lonsky (2020) and discussed by Amuedo-Dorantes and Wang (2025), are consistent with a broader shift toward forward-looking investment and self-protection. The literature overall suggests that DACA affected not only contemporaneous labour supply, but also recipients’ planning horizons, although direct evidence on how uncertainty shaped college enrolment remains comparatively scarce.

Our paper contributes to the DACA literature in three ways. First, it helps reconcile the mixed evidence on DACA’s college enrolment effects. Rather than studying DACA in isolation, we show that labour-market access and college affordability jointly shape the schooling–work allocation. Second, we tighten identification of the likely undocumented population by combining the Borjas-style residual approach with a focus on Mexican-born non-citizens, reducing likely misclassification relative to much of the earlier literature. Third, we extend the literature beyond DACA’s initial implementation by providing direct evidence on college enrolment and labour force participation during the post-2016 uncertainty period.

2.3 In-State Resident Tuition

A large first wave of research studies the extensive-margin effect of in-state resident tuition on college enrolment among likely undocumented immigrants. Early evidence is generally supportive of positive enrolment effects. Using CPS data, Kaushal (2008) finds that in-state resident tuition increased college enrolment among non-citizen Mexican young adults, and

both Flores (2010a) and Flores (2010b) report positive effects on college participation for likely undocumented Latino students. More recent CPS-based work reaches similar conclusions on average (Amuedo-Dorantes and Sparber 2014; Dickson, Gindling, and Kitchin 2017). At the same time, this literature has not reached full consensus. Most of the foundational enrolment studies use CPS data (Kaushal 2008; Flores 2010a; Amuedo-Dorantes and Sparber 2014; Dickson, Gindling, and Kitchin 2017), whereas ACS-based evidence is comparatively limited and less supportive. Using ACS supplemented by the 2000 Census, Chin and Juhn (2010) find little evidence of a statistically significant average effect on college attendance.

The literature has also branched beyond college entry to study other outcomes, including high-school enrolment and dropout, institutional choice, financing, and post-enrolment persistence and completion (Bozick and Miller 2013; Potochnick 2014; Darolia and Potochnick 2015; Amuedo-Dorantes and Sparber 2014; Conger and Turner 2017; Grosz and Hines 2022); within this broader branch, Dickson, Gindling, and Kitchin (2017) also examine the additional role of DACA and find no evidence that its introduction significantly altered the positive effect of in-state resident tuition.

A key limitation of much of the multi-state reduced-form literature is methodological. Setting aside institution-specific administrative case studies, most papers estimate staggered-adoption difference-in-differences models using two-way fixed effects. Recent econometric work shows that, under heterogeneous treatment effects across states and over time, these estimators need not recover a causal average treatment effect, motivating approaches based on group-time treatment effects and doubly robust estimation (Callaway and Sant’Anna 2021; Sant’Anna and Zhao 2020; Wooldridge 2025). Dickson, Gindling, and Kitchin (2025) address this issue directly in the in-state resident tuition context using the Callaway and Sant’Anna estimator, showing that positive average enrolment effects remain but that there is substantial heterogeneity across cohorts and implementation timing.

Our analysis builds on this literature in four ways. First, we revisit in-state resident tuition using modern staggered-adoption difference-in-differences methods designed for heterogeneous treatment effects. Second, we use ACS data over a longer horizon through 2019, allowing us to reassess the weaker ACS-based evidence. Third, we extend the outcome space beyond college enrolment to include labour force participation, allowing us to study more directly the schooling–work allocation facing undocumented immigrants. Finally, we bring the in-state resident tuition literature into closer dialogue with the DACA literature by asking whether the effect of lowering schooling costs depends on the broader policy environment governing access to work. Distinguishing between the initial DACA period and the post-2016

uncertainty period allows us to test whether uncertainty dampened the schooling benefits of in-state resident tuition by weakening the expected gains from formal labour-market access.

3 Theoretical Framework

This section develops a simple model of the higher education investment decision facing undocumented immigrants. We model the probability that higher education translates into access to higher-return formal employment, and the direct cost of entering higher education.

3.1 Schooling Choice Under Uncertain Formal-Sector Entry

Time is discrete, indexed by $t = 0, 1, 2, \dots$, and the individual discounts future utility with discount factor $\delta \in (0, 1)$. At $t = 0$, the individual chooses whether to enter higher education:

$$HE \in \{0, 1\},$$

where $HE = 1$ denotes entry into higher education and $HE = 0$ denotes non-entry.

Let within-period utility be given by a strictly increasing and strictly concave function

$$u : \mathbb{R}_+ \rightarrow \mathbb{R}, \quad u'(x) > 0, \quad u''(x) < 0.$$

The assumption of concavity ensures that the agent is risk averse. To ensure that consumption is strictly positive in all states, let $\bar{y} > c$ denote baseline resources available in each period. This can be interpreted as family support, home production, transfers, or other non-labour resources.¹ If the individual does not enter higher education, they receive wage $w_0 > 0$ immediately and in every future period. Consumption is therefore

$$c_t^N = \bar{y} + w_0 \quad \text{for all } t \geq 0.$$

The corresponding lifetime value is

$$V_0 = \sum_{t=0}^{\infty} \delta^t u(\bar{y} + w_0) = \frac{u(\bar{y} + w_0)}{1 - \delta}. \quad (1)$$

where we assume an infinite horizon. If the individual enters higher education, they incur a

¹This allows the later CRRA specification to be well defined.

direct cost $c > 0$ at $t = 0$, so current consumption is

$$c_0^H = \bar{y} - c.$$

From period $t = 1$ onward, higher education yields a risky labour-market payoff. With probability $p \in [0, 1]$, higher education leads to formal employment and wage w_1 , where $w_1 > w_0$. With probability $1 - p$, formal employment does not materialise and the individual receives the fallback wage w_0 . Thus, from $t = 1$ onward, consumption is

$$c_t^H = \begin{cases} \bar{y} + w_1 & \text{with probability } p, \\ \bar{y} + w_0 & \text{with probability } 1 - p. \end{cases}$$

For simplicity, assume this labour-market outcome is realised once and then persists forever. The lifetime value from entering higher education is therefore

$$\begin{aligned} V_1 &= u(\bar{y} - c) + \sum_{t=1}^{\infty} \delta^t [p u(\bar{y} + w_1) + (1 - p) u(\bar{y} + w_0)] \\ &= u(\bar{y} - c) + \frac{\delta}{1 - \delta} [p u(\bar{y} + w_1) + (1 - p) u(\bar{y} + w_0)]. \end{aligned} \quad (2)$$

The individual enters higher education if and only if

$$V_1 \geq V_0.$$

Define the net gain from entering higher education as

$$\Delta(p, c) \equiv V_1 - V_0.$$

Using (1) and (2),

$$\begin{aligned} \Delta(p, c) &= u(\bar{y} - c) - u(\bar{y} + w_0) + \frac{\delta}{1 - \delta} [p u(\bar{y} + w_1) + (1 - p) u(\bar{y} + w_0) - u(\bar{y} + w_0)] \\ &= u(\bar{y} - c) - u(\bar{y} + w_0) + \frac{\delta p}{1 - \delta} [u(\bar{y} + w_1) - u(\bar{y} + w_0)]. \end{aligned} \quad (3)$$

The entry rule is therefore

$$HE = \begin{cases} 1 & \text{if } \Delta(p, c) \geq 0, \\ 0 & \text{if } \Delta(p, c) < 0. \end{cases}$$

Expression (3) has a simple interpretation. The first term,

$$u(\bar{y} - c) - u(\bar{y} + w_0),$$

is the current-period utility loss from entering higher education rather than working immediately. It incorporates both the direct cost of schooling and the foregone current wage. The second term,

$$\frac{\delta}{1 - \delta} p [u(\bar{y} + w_1) - u(\bar{y} + w_0)],$$

is the discounted expected continuation gain from higher education. The probability p governs the extent to which college translates into access to the higher-return formal sector, while c governs the direct cost of entry. These are precisely the two margins of interest in the empirical analysis. In this framework, DACA can be interpreted as shifting p , while in-state resident tuition can be interpreted as shifting c .

3.2 Comparative Statics

We derive comparative statics by finding the derivative of the value difference $\Delta(p, c)$ with respect to the parameters of interest.

Differentiating (3) with respect to p gives

$$\frac{\partial \Delta(p, c)}{\partial p} = \frac{\delta}{1 - \delta} [u(\bar{y} + w_1) - u(\bar{y} + w_0)].$$

Since $w_1 > w_0$ and $u(\cdot)$ is increasing, it follows that

$$\frac{\partial \Delta(p, c)}{\partial p} > 0.$$

A higher probability that higher education leads to formal employment therefore makes entry into higher education more attractive.

Differentiating (3) with respect to c gives

$$\frac{\partial \Delta(p, c)}{\partial c} = -u'(\bar{y} - c) < 0.$$

Thus, a higher direct cost of higher education makes entry less attractive.

The implications of the model are straightforward. Policies that increase p such as DACA raise the expected return to higher education. Policies that reduce c such as in-state resident

tuition lower the current cost of entry. Both changes increase the attractiveness of higher education, but they operate through different margins.

3.3 Risk Aversion and the Uncertainty Period

The baseline framework allows for arbitrary concavity, but it is useful to impose a specific functional form in order to interpret more sharply the role of risk. To do so, suppose period utility takes the CRRA form

$$u_\gamma(x) = \begin{cases} \frac{x^{1-\gamma}}{1-\gamma}, & \gamma \neq 1, \\ \ln x, & \gamma = 1, \end{cases} \quad \gamma > 0. \quad (4)$$

The parameter γ measures relative risk aversion. In what follows, I treat an increase in γ as a parsimonious representation of the uncertainty period: when legal and political uncertainty rises, the individual behaves as if future risky returns are evaluated through a more risk-averse lens.

For notational convenience, define

$$x_c \equiv \bar{y} - c, \quad x_0 \equiv \bar{y} + w_0, \quad x_1 \equiv \bar{y} + w_1,$$

where $x_1 > x_0 > x_c > 0$ by construction. Substituting (4) into (3), the net gain from entering higher education becomes

$$\Delta_\gamma(p, c) = \frac{x_c^{1-\gamma} - x_0^{1-\gamma}}{1-\gamma} + \frac{\delta p}{1-\delta} \cdot \frac{x_1^{1-\gamma} - x_0^{1-\gamma}}{1-\gamma}.$$

for $\gamma \neq 1$, and

$$\Delta_1(p, c) = \ln x_c - \ln x_0 + \frac{\delta p}{1-\delta} [\ln x_1 - \ln x_0].$$

for $\gamma = 1$.

The comparative statics with respect to p and c are the same as in the general model:

$$\frac{\partial \Delta_\gamma(p, c)}{\partial p} = \frac{\delta}{1-\delta} [u_\gamma(x_1) - u_\gamma(x_0)] > 0,$$

and

$$\frac{\partial \Delta_\gamma(p, c)}{\partial c} = -u'_\gamma(x_c) = -x_c^{-\gamma} < 0.$$

To isolate the role of risk aversion more clearly, it is useful to work with the certainty equivalent of the higher-education continuation payoff. Let $CE_\gamma(p)$ be defined by

$$u_\gamma(CE_\gamma(p)) = p u_\gamma(x_1) + (1 - p) u_\gamma(x_0).$$

For $\gamma \neq 1$, this implies

$$CE_\gamma(p) = \left[p x_1^{1-\gamma} + (1 - p) x_0^{1-\gamma} \right]^{\frac{1}{1-\gamma}}, \quad (5)$$

while for $\gamma = 1$,

$$CE_1(p) = x_1^p x_0^{1-p}.$$

From (5), the certainty equivalent is increasing in p :

$$\frac{\partial CE_\gamma(p)}{\partial p} = \frac{1}{1 - \gamma} \left[p x_1^{1-\gamma} + (1 - p) x_0^{1-\gamma} \right]^{\frac{\gamma}{1-\gamma}} (x_1^{1-\gamma} - x_0^{1-\gamma}) > 0.$$

Thus, a higher probability that higher education leads to formal employment raises the certainty equivalent of the risky post-schooling payoff.

The certainty equivalent is also decreasing in γ for any non-degenerate lottery with $0 < p < 1$ and $x_1 \neq x_0$:

$$\frac{\partial CE_\gamma(p)}{\partial \gamma} < 0.$$

Hence, holding the payoff distribution fixed, a more risk-averse individual assigns a lower certainty equivalent to a risky prospect. Since the higher-education path is risky while the non-entry path is safe, an increase in γ lowers the attractiveness of entering higher education. In this sense, the uncertainty period can be represented conceptually as a rise in γ .

This subsection delivers a simple conceptual extension of the baseline model. In the baseline framework, a deterioration in the environment may reduce the expected return to higher education by lowering p . In the CRRA extension, the uncertainty period may also operate through a second channel: a higher effective degree of risk aversion, captured by a larger γ , lowers the certainty-equivalent value of the risky post-schooling payoff. The purpose of the model is interpretive, clarifying that periods of heightened uncertainty can weaken schooling incentives by reducing the expected return to formal employment, as well as making individuals less willing to bear risk in order to obtain that return.

4 Data

We proceed with the empirical study using American Community Survey (ACS) data, a large, nationally representative dataset that surveys about 1% of the U.S. population annually. This dataset is an individual-year level repeated cross-section, allowing us to assess outcomes related to schooling and labour force participation margins and take advantage of eligibility cut-offs for DACA and staggered adoption for in-state tuition policy. We do not observe documented status for non-citizens, so our sample will contain both documented and undocumented immigrants. This attenuates any estimates for the effect of policy that only affects undocumented immigrants towards zero, so under the SUTVA assumption, we can interpret our estimates for treatment effects as a lower-bound of the true effect on Mexican-born undocumented immigrants². We also do not observe enrolment onto DACA, so we interpret our estimates for the effect of DACA as the effect of eligibility.

To reduce the impact of this attenuation, we look at the population of “likely undocumented” Mexican-born non-citizens as a proxy for undocumented immigrants. We consider the Mexican-born population, following studies such as Kaushal (2008), Amuedo-Dorantes and Sparber (2014), and Dickson, Gindling, and Kitchin (2025) due to this population having a high percentage of undocumented immigrants. We define a non-citizen as likely undocumented using a methodology from Borjas (2017), which defines a “likely documented” non-citizen as a non-citizen who has plausibly received amnesty through IRCA³, received public benefits, served in the military, worked for the government or in a licensed occupation, or whose spouse is a citizen or likely documented. We approximate these conditions using data on year of immigration, food stamp receipt, welfare and social security income, and public health insurance receipt, veteran status, occupation codes, and spouse links⁴. This refinement reduces contamination of documented immigrants in our sample.

From the sample of likely undocumented immigrants, we define different treatment groups for each policy we study. We construct our treatment group for our DACA assessment as “likely DACA-Eligible” immigrants based on eligibility criteria established by the Department of Homeland Security. We classify a likely undocumented immigrant as likely DACA-eligible if they (1) are non-citizens who were not lawful immigrants as of June 15, 2012; (2) arrived in the U.S. before the age of 16; (3) were under the age of 31 on June 15, 2012; (4) have continuously resided in the U.S. since at least 2007; and (5) they are

²This follows the methodology of Pope (2016) and related literature.

³We define this as those who immigrated to the US before 1980.

⁴We do not use data on public health insurance for the analysis of in-state resident tuition policy since this is only observed from 2008. Removing years 2000 to 2007 would remove 6 cohorts of treated states (See section 5).

enrolled in school, have completed high school, a GED, or some college, i.e. an educational requirement. We approximate these requirements using data on non-citizens year of arrival in the US, year and quarter of birth, and highest educational qualification and school enrolment. For our assessment of in-state tuition policy, we consider treated “cohorts”, defining individuals as belonging to the same cohort if their state first offered in-state resident tuition in the same year, defined using state-specific tuition legislation.

We consider two primary outcomes, college enrolment and labour force participation. We define college enrolment as a binary, indicating if the individual is enrolled at a higher education institution. Labour force participation is defined similarly as an indicator for if an individual is working or seeking work. Following Amuedo-Dorantes and Wang (2025), we are interested in how these outcomes change for the policy periods (1) Pre-DACA (Pre-2013), (2) DACA’s Initial Implementation (2013–2016), and (3) DACA’s uncertainty period (Post-2016)⁵.

Table 1 provides summary statistics across the three policy periods for outcome variables of interest and relevant descriptive statistics for our treatment and control groups. We observe that DACA-eligible immigrants tend to have higher college and school enrolment and attainment in all periods. We also see initially lower labour force participation, and worse labour market characteristics, e.g. higher unemployment and lower wages and earnings. We also notice the large differential migration characteristics, where DACA-eligible immigrants arrived in the US at much younger ages, have been in the US for longer, and are less mobile on average. This is mechanical due to the definition of eligibility for DACA, so we control for these characteristics in our analysis. We also notice that DACA-eligible immigrants are more likely to be female, have larger families, and are less likely to be married on average. We hence control for these demographic differences in our analysis. In contrast, we notice the groups are similar in racial demographics, age, number of children, metropolitan status, and language. This provides qualitative evidence that with sufficient controls, these two groups are similar and that the control group provides a good counterfactual for our treatment group.

Table 2 provides summary statistics by in-state resident tuition status across the three policy periods. In contrast to the DACA eligibility comparison, individuals in in-state and out-of-state tuition state-periods appear broadly similar across many demographic and migration characteristics. We nevertheless observe that individuals in in-state resident tuition

⁵We define the period of uncertainty surrounding DACA as post-2016 to capture the change in the legal and political environment surrounding DACA after the 2016 presidential election and subsequent efforts to rescind the programme, as well as the general political climate surrounding undocumented immigrants. The period should be interpreted as a policy-regime classification rather than as a precise legal event date.

state-periods have higher college enrolment, school enrolment, and educational attainment in all three periods. Labour-market characteristics are also relatively similar before DACA, with near-identical rates of labour force participation and employment, although some divergence emerges during the DACA implementation period, when individuals in out-of-state tuition state-periods have higher labour force participation and employment. Migration characteristics are also much more balanced than in the DACA comparison, as individuals in the two groups have similar years of residence in the United States, age at arrival, and mobility patterns. This is important because, unlike the DACA treatment definition, in-state resident tuition exposure is not mechanically determined by individual migration characteristics. Demographic characteristics are also broadly similar across groups, including age, sex, Hispanic origin, language, marital status, family size, number of children, and metropolitan residence. The descriptive data suggests that the in-state resident tuition comparison groups are relatively similar on observable characteristics, while differing most clearly in schooling outcomes.

5 Methodology

We assess two dimensions of the decision of enrolling in higher education: (1) the return to schooling, which here takes the form of access to labour markets, and (2) the cost of schooling, i.e. tuition costs. We assess two policies that shift these dimensions and see how our sample of undocumented immigrants respond.

5.1 DACA: Methodology

We follow a similar methodology to Pope (2016) and Amuedo-Dorantes and Wang (2025), using difference-in-differences (DiD) and event study methods to estimate the average treatment effect of DACA eligibility on schooling and labour market outcomes for those likely eligible. As discussed in Section 4, we consider individuals on the margin for the higher education decision.

Identification of the intent-to-treat effect of DACA requires that, absent DACA, outcomes for DACA-eligible individuals and our control group individuals would have followed common trends in college enrolment and labour market participation conditional on controls. We justify this control group of likely undocumented Mexican-born non-citizens as a plausible counterfactual for our treatment group conditional on controls, allowing us to identify causal effects of eligibility for the policy. Conditional on migration and demographic characteristics, we argue that our treatment group would exhibit similar trends in higher ed-

Table 1: Descriptive Statistics by Group and Period

	DACA-Eligible			Control Group		
	<i>Pre-DACA</i>	<i>Implementation</i>	<i>Uncertainty</i>	<i>Pre-DACA</i>	<i>Implementation</i>	<i>Uncertainty</i>
Schooling						
In College	24.8	29.0	29.4	7.9	17.7	21.2
In School	39.1	41.4	35.9	14.2	30.4	30.6
GED	2.6	2.7	1.9	3.7	3.2	3.3
Some College	33.3	38.7	42.4	18.7	27.3	31.0
Labour market						
In Labour Force	68.2	75.7	79.0	78.2	70.7	72.0
Working	57.9	66.8	73.5	71.0	64.9	66.5
Unemployed	10.3	8.9	5.5	7.2	5.8	5.4
Wage	14,040	14,973	21,414	15,389	16,554	21,480
Earnings	13,942	14,916	21,418	15,202	16,673	21,387
Hours Worked Per Week	34.5	34.9	35.9	38.5	37.8	37.0
Worked Past Year	33.3	25.1	20.6	22.8	29.7	27.7
Self-Employed	3.2	2.4	3.5	3.3	3.3	4.3
Home Owner	56.3	58.2	56.4	81.0	71.2	65.2
Migration						
Years In US	13.1	14.7	17.6	3.5	3.8	5.3
Age Entered US	7.6	6.2	3.7	18.4	17.5	15.8
Moved In Past Year	17.1	15.0	11.4	30.7	25.6	24.9
Moved Across States	1.1	1.1	1.0	2.1	2.2	1.7
Demographics						
Male	55.1	56.4	53.2	65.3	63.4	60.0
White	60.2	64.1	42.6	59.9	66.7	29.2
Black	0.3	0.3	0.1	0.4	0.6	0.2
Asian	0.1	0.1	0.1	0.1	0.1	0.2
Hispanic	99.4	99.1	99.5	99.2	99.1	99.4
Speaks Spanish At Home	97.9	96.9	96.7	97.0	96.6	95.6
Speaks English	95.9	98.3	98.3	76.4	80.0	81.6
Age	20.7	20.9	21.4	21.9	21.3	21.1
Married	10.2	7.6	7.2	19.9	14.2	11.8
Family Size	4.6	4.4	4.4	3.6	3.8	3.9
Number Of Children	1.4	1.4	1.4	1.4	1.4	1.4
Lives In A Metro Area	88.8	88.9	88.5	85.4	86.1	88.7
N	9,969	6,529	5,742	4,138	1,828	3,993

Notes: Entries report weighted sample means from the American Community Survey for likely undocumented Mexican-born non-citizens aged 17–24 with a high school qualification or GED and no college degree. DACA-Eligible individuals satisfy the observable DACA eligibility criteria described in Section 4; the Control Group consists of otherwise comparable likely undocumented Mexican-born non-citizens who do not satisfy those criteria. The pre-DACA period is 2008–2012, the implementation period is 2013–2016, and the uncertainty period is 2017–2024. Binary variables are reported as percentages. Wages and earnings are reported in USD; hours worked are weekly hours; years in the United States, age at arrival, age, family size, and number of children are reported in levels. *N* reports the number of individual-year observations.

Table 2: Descriptive Statistics by Tuition Status and Period

	In-State Tuition			Out-of-State Tuition		
	<i>Pre-DACA</i>	<i>Implementation</i>	<i>Uncertainty</i>	<i>Pre-DACA</i>	<i>Implementation</i>	<i>Uncertainty</i>
Schooling						
In College	23.2	29.2	27.5	12.1	19.0	21.2
In School	34.8	41.3	35.2	24.2	32.3	29.0
GED	2.7	2.8	2.5	3.5	3.1	2.3
Some College	32.4	39.0	39.5	21.4	28.3	31.9
Labour market						
In Labour Force	70.7	73.4	75.9	70.4	78.2	74.3
Working	61.0	64.9	70.6	61.5	69.0	68.0
Unemployed	9.7	8.4	5.3	8.9	9.2	6.3
Wage	14,325	15,029	21,062	15,587	16,107	21,891
Earnings	14,179	15,056	21,073	15,475	15,874	21,762
Hours Worked Per Week	35.5	35.2	36.0	36.4	35.5	36.5
Worked Past Year	30.9	27.5	23.8	30.8	25.0	26.0
Self-Employed	3.2	2.5	3.7	3.4	2.6	2.1
Home Owner	62.3	61.6	61.3	66.7	61.7	53.0
Migration						
Years In US	10.7	12.4	12.5	9.5	12.5	14.0
Age Entered US	10.3	8.6	8.7	11.6	8.4	7.3
Moved In Past Year	20.5	16.6	16.2	23.5	18.2	20.2
Moved Across States	1.3	1.2	1.1	1.1	0.7	1.0
Demographics						
Male	57.3	57.3	56.3	58.2	59.9	53.4
White	59.0	66.0	37.1	67.3	60.1	51.6
Black	0.2	0.3	0.1	0.5	0.4	0.2
Asian	0.1	0.1	0.1	0.0	0.1	0.0
Hispanic	99.4	99.3	99.6	99.3	99.5	99.5
Speaks Spanish At Home	98.1	97.3	96.4	97.0	96.8	96.4
Speaks English	90.2	93.7	91.3	89.5	97.5	94.9
Age	21.1	21.0	21.2	21.1	20.9	21.3
Married	12.2	8.6	8.8	14.3	7.1	9.7
Family Size	4.5	4.4	4.2	4.1	4.2	4.2
Number Of Children	0.2	0.1	0.1	0.2	0.1	0.1
Lives In A Metro Area	92.4	90.9	91.2	87.2	93.6	94.5
N	9,634	6,303	7,483	2,341	735	553

Notes: Entries report weighted sample means from the American Community Survey for likely undocumented Mexican-born non-citizens. “In-State Tuition” denotes individuals living in a state-year in which undocumented students are eligible for in-state resident tuition under the policy coding used in the empirical analysis; “Out-of-State Tuition” denotes individuals in state-years without such access. The pre-DACA period is 2008–2012, the implementation period is 2013–2016, and the uncertainty period is 2017–2024. Binary variables are reported as percentages. Wages and earnings are reported in USD; hours worked are weekly hours; years in the United States, age at arrival, age, family size, and number of children are reported in levels. *N* reports the number of individual-year observations.

educational attainment and labour force participation. We show in Section 4 that they exhibit similar demographic characteristics. Since both groups are migrants, they would be similarly affected by US immigration policy. We show this formally in Section 6 with a discussion of non-differential pre-trends.

We estimate the following model by OLS, where we interpret β_1 and β_2 as the intent-to-treat effects of DACA eligibility on likely undocumented Mexican-born non-citizens in the implementation period and uncertainty period, relative to the base period (Pre-DACA 2008–2012):

$$\begin{aligned} Y_{ist} = & \beta_0 (\text{DACA-Eligible}_i) + \beta_1 (\text{DACA-Eligible}_i \times \text{Implementation}_t) \\ & + \beta_2 (\text{DACA-Eligible}_i \times \text{Uncertainty}_t) \\ & + X_i' \gamma + \mu_s + \tau_t + \psi_s t + \varepsilon_{ist}. \end{aligned}$$

Where DACA-Eligible_i is an eligibility indicator as described in Section 4, Implementation_t and Uncertainty_t are indicators for the given policy period, corresponding to 2013–2016 and 2017–2024 respectively, X_i is a vector of individual-level controls including age, age squared, sex, year of immigration to the US⁶⁷, and marital status, μ_s and τ_t are state and time fixed effects, and ψ_s are state-specific linear time-trends⁸. Our parameters of interest are β_1 and β_2 . Standard errors are clustered at the state level, allowing for arbitrary within-state correlation and heteroskedasticity in the error terms, including serial correlation over time. Since we use repeated cross-sectional data, we use person weights for each observation⁹ We estimate the regression separately for outcomes Y_{ist} college enrolment and labour force participation, where we interpret this as a linear probability model.

We similarly estimate an event study by OLS to assess pre-trends to find supporting evidence for the common trends assumption, and assess finer dynamics of the effect of DACA, using 2012 as the reference year, as this was the final year before DACA was fully implemented:

$$\begin{aligned} Y_{ist} = & \delta \text{DACA-Eligible}_i + \sum_{\tau \neq 2012} \beta_\tau (\text{DACA-Eligible}_i \times \mathbf{1}\{t = \tau\}) \\ & + X_i' \gamma + \mu_s + \tau_t + \psi_s t + \varepsilon_{ist}. \end{aligned}$$

⁶Year of immigration to the US in 5 year bins.

⁷Age at immigration to the US is a linear combination of year of immigration to the US, year, and age, hence we do not include this due to collinearity.

⁸Amuedo-Dorantes and Sparber (2014) show sensitivity of results to state-specific linear time-trends in related studies.

⁹Using the ACS variable PERWT.

We perform robustness checks by relaxing the control group, considering Mexican-born non-citizens, likely undocumented immigrants, and non-citizens within the higher education choice margin, and we report the estimates in Figure 2.

5.2 In-State Resident Tuition: Methodology

To study the schooling-cost margin, we exploit staggered state adoption of eligibility for in-state resident tuition for undocumented immigrants following a similar methodology to Dickson, Gindling, and Kitchin (2025). These policies reduce the direct cost of college attendance and therefore provide a natural policy shock to the net returns to higher enrolment for undocumented immigrants. Table 3 reports the first year in which each state offered in-state resident tuition to undocumented immigrants and to DACA recipients. We similarly conduct our analysis on the sample of likely undocumented Mexican-born non-citizens, where treatment is defined on the state-level. We assess how the effects of in-state resident tuition differ across the pre-DACA, implementation, and uncertainty periods.

Because states adopted in-state resident tuition at different points in time, we estimate treatment effects using the staggered-adoption difference-in-differences framework of Callaway and Sant’Anna (2021)¹⁰. Let $G_i = g$ denote the first year in which individual i ’s state of residence offers in-state resident tuition to the relevant treatment group, and let $G_i = 0$ for individuals in never-treated states. For each treated cohort g and calendar year $t \geq g$, the group-time average treatment effect is

$$ATT(g, t) = \mathbb{E}[Y_{it}(g) - Y_{it}(0) \mid G_i = g],$$

where $Y_{it}(g)$ denotes the potential outcome for individual i in year t if first exposed to treatment in year g , and $Y_{it}(0)$ denotes the potential outcome in the absence of treatment. We then estimate and aggregate these treatment effects for each year across cohorts using the weighting scheme in Callaway and Sant’Anna (2021), and then aggregate the calendar-year treatment effects with equal weighting within the pre-DACA, implementation, and uncertainty periods to obtain policy-period specific treatment effects¹¹.

We estimate these treatment effects using repeated cross-sections and variation over states and time. In each year, treated individuals are compared with individuals living in states that are either never treated or not yet treated by that year. This comparison is appropriate under a conditional parallel trends assumption: absent the policy, average

¹⁰We use these methods to address bias created under staggered-treatment and heterogeneous effects, discussed in Chaisemartin and D’Haultfœuille (2020) and Goodman-Bacon (2021).

¹¹We use the `lincom` command in Stata for this aggregation.

Table 3: First Treatment Year for In-State Resident Tuition Policies

State	Undocumented Immigrants	DACA Recipients
Arizona	2023	2023
Arkansas	—	2019
California	2002	2002
Colorado	2013	2013
Connecticut	2011	2011
Florida	2014	2014
Hawaii	2013	2013
Illinois	2003	2003
Kansas	2004	2004
Kentucky	2015	2015
Maryland	2013	2013
Massachusetts	2023	2013
Minnesota	2013	2013
Nebraska	2006	2006
Nevada	2021	2021
New Jersey	2014	2014
New Mexico	2005	2005
New York	2002	2002
Ohio	—	2013
Oklahoma	2003	2003
Oregon	2013	2013
Rhode Island	2012	2012
Texas	2001	2001
Utah	2002	2002
Virginia	2020	2014
Washington	2003	2003

Notes: Entries report the first year in which each state is coded as offering in-state resident tuition under each treatment-timing definition used in the paper. “Undocumented Immigrants” uses the first year of broader access for undocumented students; “DACA Recipients” uses the first year in which DACA recipients are eligible under state policy. States not listed are never treated during the sample period. “—” indicates that the state is untreated under that definition during the sample period. If a policy became effective after September 1, treatment is coded as beginning in the following calendar year.

outcomes for treated and comparison units with the same observed characteristics would have evolved similarly over time. Following Dickson, Gindling, and Kitchin (2025), we also impose a no-anticipation assumption, so that outcomes are unaffected before a state’s in-state resident tuition policy becomes available. We claim this is plausible as anticipation cannot affect the cost of college before implementation.

Our preferred specification uses the doubly robust estimator to recover the counterfactual outcome path for treated units, that is, the outcomes those units would have experienced in the absence of treatment. It does so using two adjustment strategies: outcome regression, which models the outcome over time as a function of observables, and inverse-probability weighting, which reweights comparison units so that they resemble treated units on observed characteristics. The estimated treatment effect is then given by the difference between the observed outcome path of treated units and this estimated counterfactual outcome path. This estimator is attractive because it is consistent if either the propensity-score model or the outcome-regression model is correctly specified, rather than requiring both to be correct. We use the same set of control variables X_i for these adjustment strategies, including age, age squared, sex, marital status, race, and year of immigration to the US.

All estimates use person weights, and standard errors are clustered at the state level. The person weights make the estimates representative of the relevant population in each repeated cross-section, while state-level clustering allows inference to be robust to arbitrary heteroskedasticity and correlation in the regression errors among observations within the same state, including correlation across years.

6 Results

In this section we discuss the results for the estimates from the models discussed in Section 5. We will discuss (1) the difference-in-differences ITT estimates for the likely *DACA-Eligible* Mexican-born non-citizen group and (2) the ATT estimates for likely undocumented Mexican-born non-citizens of in-state resident tuition.

6.1 DACA: Results

Table 4 shows that DACA’s initial implementation increased labour force participation among likely eligible immigrants, with little evidence of a contemporaneous change in college enrolment. The coefficient on *DACA-Eligible* indicates that, before DACA, eligible individuals were 5.7 percentage points more likely to be enrolled in college than ineligible individuals, while the pre-DACA difference in labour force participation was small and statistically in-

significant. The interaction terms then estimate how these outcomes changed for eligible individuals during each DACA period relative to the pre-DACA baseline.

During the implementation period, DACA eligibility is associated with a 7.1 percentage point increase in labour force participation, significant at the 0.1 percent level. By contrast, the corresponding estimate for college enrolment is -2.8 percentage points and is not statistically distinguishable from zero. The implementation-period estimates therefore suggest that DACA’s initial effect operated primarily through the labour-market margin rather than through increased college enrolment.

The uncertainty-period estimates point to a weaker labour-market response and a decline in college enrolment. Relative to the pre-DACA period, DACA eligibility is associated with a 4.5 percentage point increase in labour force participation, smaller than the implementation-period estimate, and significant at the 5 percent level. For college enrolment, the uncertainty-period estimate is -6.5 percentage points and significant at the 1 percent level. Thus, once DACA entered a more uncertain policy environment, eligible immigrants appear to have experienced lower college attachment alongside attenuated labour-market gains.

Overall, Table 4 suggests that DACA initially shifted likely eligible immigrants toward labour force participation without increasing college enrolment. In the later uncertainty period, the labour-force response weakened and college enrolment fell, consistent with policy instability reducing longer-term educational investment while partially preserving the labour-market effects of work authorization.

Appendix Tables 6 and 7 show that the DACA estimates are sensitive to specification choices that account for baseline differences between eligible and ineligible individuals. For both college enrolment and labour force participation, the estimates change materially once state and year fixed effects are included, and again after adding age controls and year-of-immigration controls. By contrast, adding state-specific linear trends has little effect, suggesting that the baseline results are not driven by simple differential state trends. The labour-force estimates are also somewhat sensitive to sex controls, which account for part of the pre-DACA difference in participation between eligible and ineligible individuals, although the positive implementation-period response remains. Additional controls for marital status and race have little impact. Overall, the specification checks suggest that age structure and immigration cohort are important confounders, while the main qualitative results are preserved in the preferred fully controlled specification.

Figure 1 and Appendix Table 8 report event-study estimates using 2012 as the omitted reference year. The estimates provide little evidence of systematic differential pre-trends

Table 4: College Enrolment and Labour Force Participation Responses to DACA

	In College	In Labour Force
DACA-Eligible	5.6618*** (1.6109)	-1.7417 (1.1487)
DACA-Eligible $\times$ Implementation	-2.7928 (1.7713)	7.0640*** (1.5745)
DACA-Eligible $\times$ Uncertainty	-6.4683** (1.9989)	4.4883* (1.7717)
Age (+ Age Squared)	Y	Y
Sex	Y	Y
Marital Status	Y	Y
Race	Y	Y
Year of Immigration	Y	Y
State FE	Y	Y
Year FE	Y	Y
State-specific trends	Y	Y
Clustered SE	State	State
N	32,199	32,199

Notes: Entries are linear probability model coefficients reported in percentage points. The sample consists of likely undocumented Mexican-born non-citizens in the American Community Survey on the college enrolment margin described in Section 4. “DACA-Eligible” identifies individuals satisfying the observable DACA eligibility criteria; because DACA receipt is not observed, estimates should be interpreted as intent-to-treat effects of DACA eligibility. The row “DACA-Eligible” reports the baseline difference between likely eligible and ineligible individuals in the pre-DACA period. The interaction rows report changes for likely DACA-eligible individuals relative to likely ineligible individuals and relative to the pre-DACA baseline. The implementation period is 2013–2016 and the uncertainty period is 2017–2024. All specifications include controls for age, age squared, sex, marital status, race, and year of immigration, as well as state fixed effects, year fixed effects, and state-specific linear time trends. Standard errors are in parentheses and are clustered at the state level. $\cdot p < 0.1$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

before DACA. The pre-2012 coefficients are generally small and statistically insignificant, with only limited evidence of a marginal pre-period difference for college enrolment in 2008. After DACA, the dynamics are consistent with the baseline results. Labour force participation rises most clearly in the years following implementation, before weakening later in the sample, while college enrolment becomes increasingly negative during the uncertainty period. Overall, the event-study evidence supports the parallel-trends assumption and suggests that DACA’s labour-market effects were strongest shortly after implementation, whereas the decline in college enrolment emerged more gradually under policy uncertainty.

6.2 In-State Resident Tuition: Results

Table 5 shows that in-state resident tuition increases college enrolment among likely undocumented immigrants, while reductions in labour force participation are concentrated in the DACA implementation and uncertainty periods. The overall ATT for college enrolment is 4.2 percentage points and statistically significant at the 5 percent level. The corresponding overall labour-force estimate is negative, at -2.2 percentage points, but is not statistically distinguishable from zero. The regime-specific estimates therefore suggest that lower tuition consistently raises college enrolment, but affects labour supply mainly once the broader DACA environment changes the returns to formal work.¹²

Before DACA, in-state resident tuition increases college enrolment modestly, with little evidence of a corresponding labour-force response. The pre-DACA ATT for college enrolment is 3.6 percentage points and marginally significant, while the labour-force estimate is small, positive, and statistically insignificant. Thus, before DACA expanded access to formal employment, lower tuition appears to operate primarily through the direct schooling-cost margin rather than through a clear reallocation away from work.

During the DACA implementation period, in-state resident tuition produces a stronger two-sided response, raising college enrolment and reducing labour force participation. The college-enrolment ATT rises to 6.8 percentage points and is statistically significant at the 1 percent level. At the same time, the labour-force ATT is -7.8 percentage points and statistically significant at the 5 percent level. These estimates suggest that when DACA made formal employment more accessible, reducing college costs shifted likely undocumented immigrants toward schooling and away from labour force participation.

In the uncertainty period, in-state resident tuition continues to increase college enrolment while the negative labour-force response persists. The college-enrolment ATT is 4.3

¹²Appendix Table 15 reports the underlying calendar-year ATTs. The regime-specific estimates in Table 5 aggregate these annual effects across the pre-DACA, implementation, and uncertainty periods.

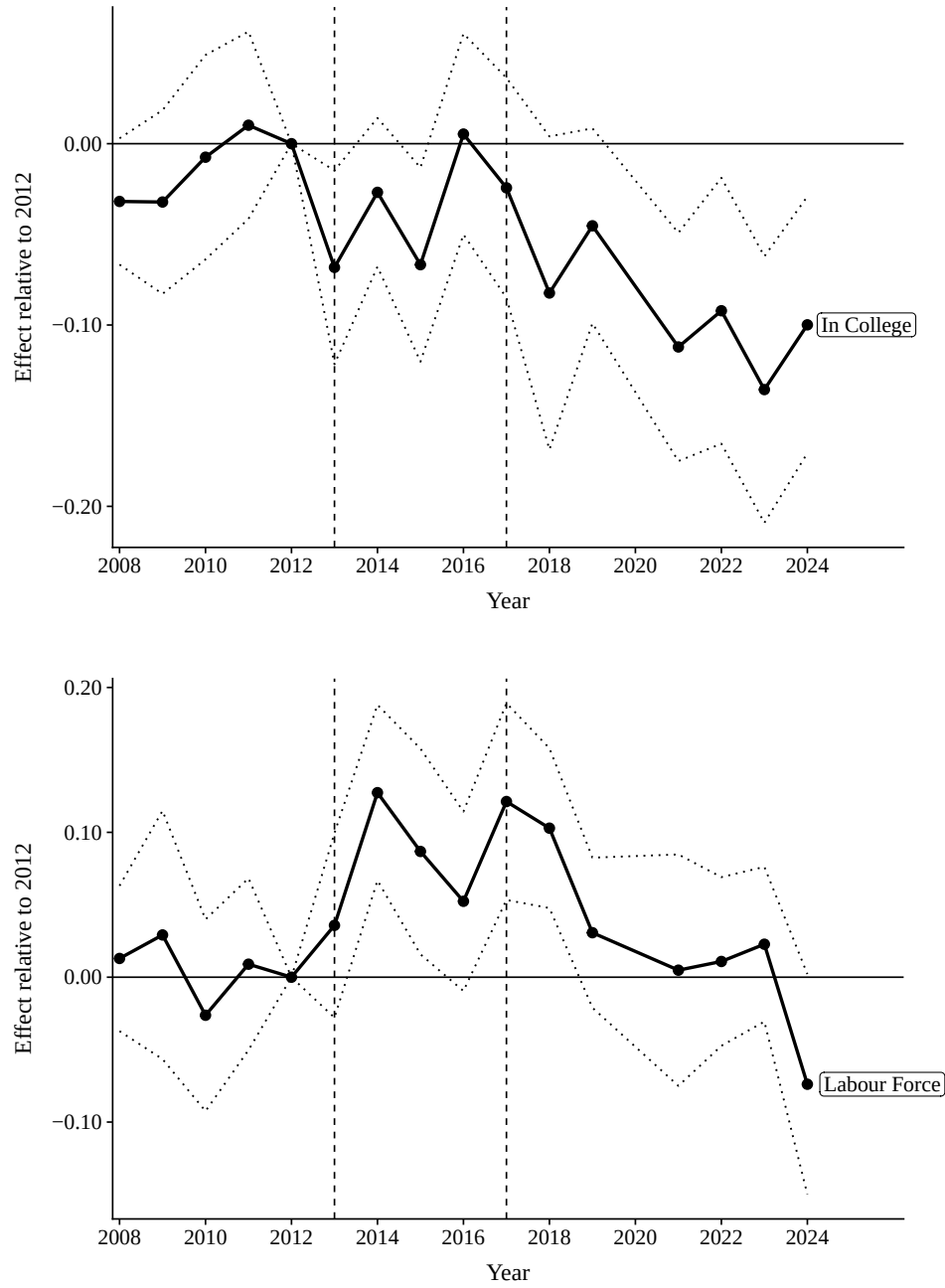


Figure 1: DACA event-study estimates for college enrolment and labour force participation.

Notes: The upper panel reports estimates for college enrolment and the lower panel reports estimates for labour force participation. Points plot event-study coefficients from a difference-in-differences specification comparing likely DACA-eligible individuals with the control group, using 2012 as the omitted reference year. Estimates are reported in probability units, so 0.10 corresponds to 10 percentage points. Dotted lines report 95 percent confidence intervals. The first vertical dashed line marks the beginning of the DACA implementation period, and the second vertical dashed line marks the beginning of the uncertainty period. The sample consists of likely undocumented Mexican-born non-citizens in the American Community Survey. Specifications include demographic controls, state fixed effects, year fixed effects, and state-specific linear trends, with standard errors clustered at the state level.

percentage points and statistically significant at the 10 percent level, while the labour-force ATT is -8.3 percentage points and statistically significant at the 1 percent level. Relative to the implementation-period estimates, the point estimate for college enrolment is smaller, while the labour-force effect remains large. This pattern suggests that policy uncertainty weakened the schooling response to lower tuition, but did not eliminate the reallocation away from work.

The table indicates that the effect of in-state resident tuition depends on the surrounding DACA policy regime. Before DACA, lower tuition raises college enrolment without a clear labour-force effect; during implementation, it raises enrolment and reduces labour force participation; and under uncertainty, the schooling effect remains positive but smaller while the labour-force reduction persists. These results are consistent with a schooling–labour trade-off that becomes most visible when lower college costs interact with expanded formal work opportunities under DACA.

Appendix Tables 10–11 and Appendix Figures 4a–4b report event-study estimates around the first year of in-state resident tuition eligibility. The main purpose of these estimates is to assess whether treated and comparison units were already evolving differently before treatment. Taken as a whole, the pre-treatment estimates provide supporting evidence against systematic differential trends: the average pre-treatment ATT is close to zero and statistically insignificant for both college enrolment and labour force participation, even though some individual lead estimates are noisy. The post-treatment effects broadly mirror the aggregated results, with college enrolment tending to rise and labour force participation tending to fall after eligibility.

Overall, the results show that undocumented immigrants respond to both policies, but along different margins. DACA initially increases labour force participation among likely eligible immigrants, with little evidence of a contemporaneous rise in college enrolment. In the later uncertainty period, these labour-market gains weaken and college enrolment falls. In-state resident tuition, by contrast, increases college enrolment throughout, but its labour-force effects depend on the DACA regime: before DACA, lower tuition raises enrolment without significantly affecting labour force participation, whereas after DACA implementation it raises college enrolment and reduces labour force participation. These results jointly suggest that undocumented immigrants respond along both the return-to-schooling margin and the cost-of-schooling margin, and that work authorisation creates a sharper trade-off between schooling and work.

Table 5: College Enrolment and Labour Force Participation Responses to In-State Resident Tuition for Undocumented Immigrants by DACA Policy Regime

	<i>In College</i>	<i>In Labour Force</i>
ATT	4.1822** (1.5127)	-2.1675 (1.5329)
ATT (Pre-DACA)	3.6152 (1.9031)	0.8115 (1.7401)
ATT (Implementation)	6.7794*** (1.7835)	-7.7614** (2.6913)
ATT (Uncertainty)	4.3361* (2.0448)	-8.3468*** (2.0645)
Age (+ Age Squared)	Y	Y
Sex	Y	Y
Marital Status	Y	Y
Race	Y	Y
Year of Immigration	Y	Y
N	68987	68987

Notes: Entries report average treatment effects on the treated (ATTs) from the Callaway and Sant’Anna staggered-adoption difference-in-differences estimator for repeated cross-sections, reported in percentage points. The sample consists of likely undocumented Mexican-born non-citizens in the American Community Survey on the college enrolment margin described in Section 4. Treated individuals are those living in states that offer in-state resident tuition to undocumented immigrants; comparison units are individuals in never-treated or not-yet-treated states. The regime-specific rows report calendar-time aggregated ATT estimates within each DACA policy regime, while the overall row reports the simple aggregated ATT across post-treatment periods. The pre-DACA period covers 2001–2012, the implementation period covers 2013–2016, and the uncertainty period covers 2017–2019. Post-2019 years are excluded from the preferred estimates because they coincide with COVID-era disruptions. Standard errors are reported in parentheses and clustered at the state level. $\cdot p < 0.1$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

7 Discussion

In this section we (1) discuss the findings in this paper, (2) the policy implications, and (3) the limitations of our empirical analysis.

7.1 Interpretation

The theoretical framework in Section 3 predicts that the schooling decision depends on the net value of higher education, denoted by $\Delta(p, c)$. Individuals enrol in college when the expected value of higher education exceeds the value of not enrolling. In this framework, DACA raises the expected return to schooling by increasing the probability that human capital can be used in the formal labour market, while in-state resident tuition raises the value of college by lowering its direct cost. The empirical estimates should therefore be interpreted as shifts in the share of individuals for whom $\Delta(p, c)$ crosses the enrolment threshold, rather than as mechanical responses by all eligible individuals.

We interpret the effects of DACA under this framework. During the implementation period, DACA substantially increases labour force participation, but we find little evidence of a contemporaneous increase in college enrolment. We interpret this as DACA providing an insufficient shift in $\Delta(p, c)$ toward the college enrolment threshold. If DACA raises the value of higher education, but only moves a limited mass of individuals across the schooling threshold, the average enrolment response may be small or statistically imprecise. The insignificant college-enrolment effect is thus consistent with a weak or offsetting net shift in $\Delta(p, c)$ at the higher education margin. By contrast, in-state resident tuition operates directly on the cost of college. Its positive effect on enrolment is therefore the clearest evidence that undocumented students respond to the cost-of-schooling margin.

The strongest evidence in the paper comes from the interaction between these two margins. Once DACA is in place, the effect of in-state resident tuition on college enrolment becomes larger, and the labour-force effect turns negative. This pattern is consistent with the model because the two policies shift $\Delta(p, c)$ in the same direction: DACA raises the expected return to acquiring human capital, while in-state resident tuition lowers the cost of doing so. In a threshold model, these policies need not have interaction effects at the individual level to generate larger aggregate enrolment responses. If DACA moves more individuals close to the college-enrolment margin, then a reduction in tuition can induce a larger share of them to cross the threshold into college. The evidence therefore suggests that tuition subsidies matter most when undocumented students can expect their schooling investments to be rewarded in the formal labour market.

The uncertainty period operates through a separate channel from the return-to-schooling and cost-of-schooling mechanisms captured by DACA and in-state resident tuition. In the theoretical framework, this channel is represented by a fall in the certainty equivalent of future payoffs, modelled as an increase in the CRRA parameter γ , where individuals behave as if future formal-sector returns have become riskier, making both work and long-horizon schooling investments less attractive. The empirical estimates align with this prediction, as the post-2016 DACA estimates show negative effects on both college enrolment and labour force participation. This pattern is consistent with DACA-eligible individuals being differentially exposed to policy uncertainty, producing a bigger change in γ . Unlike ineligible individuals, they had access to work authorisation and deportation relief whose continuation became uncertain. Formally, if DACA raises the expected return to schooling through a higher probability p of accessing formal-sector opportunities, then uncertainty about DACA's continuation lowers the certainty equivalent of those future returns. The labour-force result can be read as an attenuation of DACA's initial employment benefits, while the college-enrolment result reflects a direct negative effect of uncertainty on schooling investment.

The in-state tuition results during the uncertainty period support this interpretation. Tuition reductions continue to raise college enrolment, although the effect is smaller than during DACA implementation. This pattern suggests that lowering the direct cost of college remains effective even when the expected return to schooling becomes more uncertain. However, uncertainty appears to weaken the environment in which cost reductions translate into enrolment. Overall, the results indicate that undocumented immigrants respond to both the return-to-schooling and cost-of-schooling margins, but that these responses depend on whether legal status is perceived as stable enough for schooling investments to pay off.

7.2 Policy Implications

The main policy implication is that work authorisation and college affordability should not be evaluated in isolation. DACA expands access to formal employment, but the estimates show little evidence that work authorisation alone increases college enrolment. In contrast, in-state resident tuition increases college enrolment, and the effect is largest during the period in which DACA also expands access to formal work. This suggests that policies increasing the return to schooling may be more effective when paired with policies lowering the cost of schooling. If policymakers aim to increase college enrolment and the skilled labour supply among undocumented immigrants, work authorisation alone may be insufficient. Tuition subsidies, access to state financial aid, or policies relaxing short-run credit constraints may be important complements, although a full cost-benefit analysis would need to account for

fiscal costs, completion rates, future earnings, tax contributions, and spillovers to other students. This also speaks to the general evaluation of policies affecting schooling choices, as they may interact with other policy.

The results also suggest that legal and political uncertainty can itself affect economic behaviour. During the uncertainty period, the labour-market gains from DACA weaken and college enrolment falls among likely eligible immigrants. In the framework of this paper, uncertainty lowers the value of long-horizon investment by reducing the expected payoff from formal-sector access and by making the risky schooling path less attractive. The policy implication is that the durability of legal status matters for the effectiveness of both labour-market and education policies. Temporary or contested work authorisation may expand employment in the short run, but it may be less effective at encouraging investment in multi-year schooling if recipients doubt that the benefits will persist. Low-cost informational interventions may reduce uncertainty about eligibility rules, application procedures, or tuition access, but they are unlikely to substitute fully for stable legal and institutional design.

We interpret the effect of in-state resident tuition on labour force participation. The negative response should be interpreted as a short-run time-allocation response, not necessarily as a welfare loss. Once DACA makes formal work more accessible, lowering the cost of college raises enrolment and reduces labour force participation. This pattern is consistent with individuals reallocating time away from work and toward schooling when the expected return to education rises and the direct cost of enrolment falls. If the schooling investment leads to higher completion, better job matching, or higher future earnings, then lower contemporaneous labour force participation may be the short-run cost of higher future human capital. The estimates in this paper cannot test that long-run channel directly. Further research should therefore examine whether the enrolment gains induced by in-state tuition translate into degree completion, occupational mobility, and earnings growth.

These findings are directly relevant to recent reversals of in-state resident tuition access in Texas, Florida, and Oklahoma. The estimates in this paper predict that such reversals in in-state tuition access should reduce college enrolment among affected undocumented students, all else equal. They also suggest a broader uncertainty effect, where even where formal eligibility remains, perceived instability in tuition policy may weaken the enrolment response if students become less confident that the policy will persist over the relevant schooling period. These predictions should be interpreted as extrapolations from the estimated historical relationships, not as a complete evaluation of recent state policy changes. The results identify behavioural responses along the schooling and labour margins, rather than

the full welfare, fiscal, or legal consequences of expanding or removing tuition access.

7.3 Limitations

A first identification concern in both the DACA DiD model and the in-state resident tuition CSDiD model is the credibility of conditional parallel trends. Our event-study evidence on pre-trends is supportive, but it cannot verify the identifying assumption, since parallel trends is inherently counterfactual. In the DACA setting, this concern is especially important as related work has emphasized that DACA eligibility rules can themselves create systematic differences between treated and control groups, making counterfactual comparability non-trivial. In addition, the repeated cross-sectional nature of the data raises a compositional concern. The share and quantity of likely DACA-eligible observations fall over time, so later-period estimates may partly reflect selection into or out of the sample rather than pure behavioural responses. A further issue is endogeneity of controls, as marital status is best viewed as a potentially post-treatment variable, and therefore a possible bad control, because DACA may itself affect family formation and intermarriage decisions, particularly during the uncertainty period.

The same identification logic applies to the in-state resident tuition analysis, but with additional concerns specific to the staggered-adoption CSDiD design. The Callaway–Sant’Anna framework still relies on conditional parallel trends and no anticipation, even though it avoids the weighting problems of standard staggered two-way fixed effects. In practice, however, undocumented immigrants may select into states with more favourable educational environments, or may move in anticipation of lower tuition costs, better public services, or broader immigrant-friendly institutions. If individuals systematically sort into “nicer” states on dimensions related to college enrolment or labour force participation, then treated and control states may differ in ways that are not fully absorbed by observed covariates. This same sorting logic also weakens the no-anticipation assumption, as even before formal implementation, the prospect of future in-state tuition may alter expectations about the returns to schooling relative to work, changing behaviour in untreated periods. Treatment timing is also imperfectly measured, since actual eligibility depends on satisfying state-specific residency and high-school criteria, not simply current residence in a treated state. These issues imply that both models should be read as credible only to the extent that observed controls successfully proxy for the main sources of differential trend formation, and that neither design can fully rule out residual bias from unobserved selection, endogenous controls, or anticipatory behaviour.

A further limitation concerns the construction of the likely undocumented sample

itself. Our filter is designed to improve internal validity by removing non-citizens who are likely legally present, but some of the variables used in that screen may themselves respond to DACA. For example, in the Borjas-style residual method, individuals are classified as likely documented if, among other conditions, they work in a licensed occupation or have a spouse who is a legal immigrant or citizen. This creates a potential source of endogenous sample selection in the DACA analysis, as if DACA improves occupational mobility or labour market access, treated individuals may become more likely to enter licensed work, and if DACA or subsequent uncertainty affects marriage decisions, eligible immigrants may also become more likely to marry citizens or likely documented non-citizens. If this occurs, some of the most strongly affected DACA-eligible individuals are mechanically filtered out of the post-treatment sample, which would tend to attenuate estimated DACA effects, especially on labour force participation. At the same time, restricting attention to Mexican-born non-citizens improves internal validity by reducing contamination from legal immigrants and focusing on a group with a high undocumented share, but it narrows the target population. The estimates therefore speak most directly to likely undocumented Mexican-born non-citizens rather than undocumented immigrants more generally, so the gain in internal validity comes at the cost of external validity.

We discuss some further data and modelling limitations. Some of the in-state resident tuition estimates are built on very thin support. In the Callaway–Sant’Anna framework, identification proceeds through cohort-time treatment effects, but in our setting many cohort-year cells are small, so some effects are estimated from very little data and may rely on weak overlap between treated and untreated observations. This problem is made more severe by the undocumented filter, which reduces misclassification of undocumented status at the cost of shrinking already small cells, and it would become still more acute if we further restricted the in-state resident tuition analysis to the DACA-eligible population, which is why we do not estimate that specification. A final issue is functional form. As outcomes are binary, the linear probability model imposes a constant partial effect of treatment on the probability of enrolment or labour force participation, which may be unrealistic if policy effects differ across individuals or along the probability distribution. The coefficients should therefore be interpreted as approximate percentage-point changes in probability.

8 Conclusion

This paper studies how undocumented immigrants allocate between college enrolment and labour force participation when policy changes relax different constraints on schooling and work. Using repeated cross-sections from the American Community Survey, we estimate the

effects of DACA eligibility through difference-in-differences and event-study methods, and the effects of in-state resident tuition through a staggered-adoption difference-in-differences design.

The results show that undocumented immigrants respond to both the return to schooling and the cost of schooling, but that these responses are best understood jointly. DACA initially increases labour force participation among likely eligible immigrants, with little evidence of a contemporaneous rise in college enrolment. During the later uncertainty period, these labour-market gains weaken and college enrolment falls. In-state resident tuition, by contrast, increases college enrolment throughout, but its effects depend on the surrounding DACA regime. Before DACA, lower tuition raises enrolment without a clear labour-force response. Once DACA expands access to formal work, the enrolment response to in-state tuition strengthens and labour force participation falls. These patterns suggest a schooling–labour trade-off: when formal work becomes more accessible and college becomes less costly, some individuals reallocate time away from work and toward higher education.

The paper contributes to the literature in three ways. First, it helps reconcile mixed findings on DACA and in-state resident tuition by considering both margins in the same framework. DACA and tuition policy do not operate independently: work authorisation changes the expected return to human capital, while in-state tuition changes the cost of acquiring it. Studying either policy in isolation can therefore obscure the trade-off between schooling and work. Second, the paper provides a theoretical framework for this setting, showing how the value of higher education depends on both expected formal-sector access and direct schooling costs. This framework clarifies why the same tuition policy can have different effects across DACA regimes. Third, the paper studies the DACA uncertainty period directly. The results suggest that legal and political instability can weaken the effects of work authorisation and reduce investment in college, consistent with the idea that temporary or contested legal status shortens planning horizons and lowers the value of long-run schooling investments.

These findings also speak to current immigration and education policy. Recent reversals of in-state resident tuition access in Texas, Florida, and Oklahoma are likely to reduce college enrolment among affected undocumented students, all else equal. The estimates in this paper do not provide a complete welfare analysis of those policy changes, but they imply that raising the direct cost of college should move some students away from enrolment. Furthermore, the results suggest that legal uncertainty can reduce the effectiveness of both labour-market and education policy: Work authorisation may increase labour force participation in the short run, but its ability to encourage longer-run investment depends on

whether recipients expect the benefits to persist. Similarly, tuition access may have weaker effects when students doubt that eligibility rules will remain stable over the period required to complete college.

Future research should study the mechanisms and long-run consequences behind these short-run responses. One direction is to examine schooling costs in more detail, including access to financial aid, student loans, and variation in the size of tuition subsidies. This would help distinguish whether undocumented students respond mainly to posted tuition, liquidity constraints, or the horizon in which education pays off. A second direction is to move beyond enrolment and study other margins of college investment, including persistence, completion, academic performance, and the individual wage premium to higher education. Administrative or linked education–earnings data would be especially useful for this purpose. A third direction is to assess whether the labour-force decline associated with in-state tuition is temporary. If lower labour force participation leads to higher completion, better job matching, or higher future earnings, then the short-run reduction in work may represent investment rather than withdrawal. Finally, future work should study heterogeneity across demographic groups, states, local labour markets, and migration histories. These differences are likely to matter for both the constraints undocumented immigrants face and the policies most likely to relax them.

Overall, the evidence points to the central lesson that policies that affect undocumented immigrants’ schooling and labour decisions should not be evaluated one margin at a time. Work authorisation, tuition access, and policy certainty jointly shape the returns, costs, and risks of higher education. Understanding that interaction is essential for explaining past responses to DACA and in-state resident tuition, and for evaluating future policies aimed at the economic integration of undocumented immigrants.

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9 Appendix

Table 6: College Enrolment Responses to DACA Across Specifications

	<i>Model</i>							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
DACA-Eligible	9.8124*** (1.8957)	14.5831*** (2.1163)	14.3584*** (2.1993)	11.9697*** (2.1717)	11.3554*** (2.1123)	10.6019*** (2.1628)	10.6083*** (2.1758)	5.6618*** (1.6109)
DACA-Eligible $\times$ Implementation	4.2146*** (0.7455)	-3.4604 (1.7839)	-3.2477 (1.8452)	-2.1730 (1.7391)	-2.0284 (1.7657)	-2.0069 (1.7996)	-2.0181 (1.8018)	-2.7928 (1.7713)
DACA-Eligible $\times$ Uncertainty	4.5680** (1.3231)	-7.8612*** (1.6532)	-7.5155*** (1.8251)	-4.2587* (1.6352)	-4.1724* (1.6300)	-4.3407* (1.6539)	-4.3073* (1.6634)	-6.4683** (1.9989)
Age (+ Age Squared)				Y	Y	Y	Y	Y
Sex					Y	Y	Y	Y
Marital Status						Y	Y	Y
Race							Y	Y
Year of Immigration								Y
State FE		Y	Y	Y	Y	Y	Y	Y
Year FE		Y	Y	Y	Y	Y	Y	Y
State-specific trends			Y	Y	Y	Y	Y	Y
Clustered SE	State	State	State	State	State	State	State	State
N	32,199	32,199	32,199	32,199	32,199	32,199	32,199	32,199

Notes: Entries are linear probability model coefficients for college enrolment, reported in percentage points. The sample consists of likely undocumented Mexican-born non-citizens in the American Community Survey on the college enrolment margin described in Section 4. “DACA-Eligible” identifies individuals satisfying the observable DACA eligibility criteria; because DACA receipt is not observed, estimates are intent-to-treat effects of eligibility. The row “DACA-Eligible” reports the baseline pre-DACA difference between likely eligible and ineligible individuals. The interaction rows report period-specific changes for likely DACA-eligible individuals relative to likely ineligible individuals and relative to the pre-DACA baseline. Models add fixed effects, state-specific linear trends, and demographic controls cumulatively as indicated. Standard errors are in parentheses and are clustered at the state level. $\cdot p < 0.1$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

Table 7: Labour Force Participation Responses to DACA Across Specifications

	<i>Model</i>							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
DACA-Eligible	-6.1004*** (0.8824)	-8.8646*** (1.1069)	-8.7654*** (1.1241)	-2.2209 (1.1486)	-0.7843 (1.0863)	-1.4160 (1.0259)	-1.4530 (1.0173)	-1.7417 (1.1487)
DACA-Eligible $\times$ Implementation	7.4540*** (1.2117)	14.1128*** (1.3784)	14.0602*** (1.4097)	8.7800*** (1.3955)	8.4417*** (1.3305)	8.4597*** (1.3388)	8.4697*** (1.3276)	7.0640*** (1.5745)
DACA-Eligible $\times$ Uncertainty	10.7677*** (0.9963)	15.8306*** (1.8433)	15.7431*** (1.8861)	6.9916*** (1.8080)	6.7897*** (1.8118)	6.6487*** (1.8014)	6.6710*** (1.7891)	4.4883* (1.7717)
Age (+ Age Squared)				Y	Y	Y	Y	Y
Sex					Y	Y	Y	Y
Marital Status						Y	Y	Y
Race							Y	Y
Year of Immigration								Y
State FE		Y	Y	Y	Y	Y	Y	Y
Year FE		Y	Y	Y	Y	Y	Y	Y
State-specific trends			Y	Y	Y	Y	Y	Y
Clustered SE	State	State	State	State	State	State	State	State
N	32,199	32,199	32,199	32,199	32,199	32,199	32,199	32,199

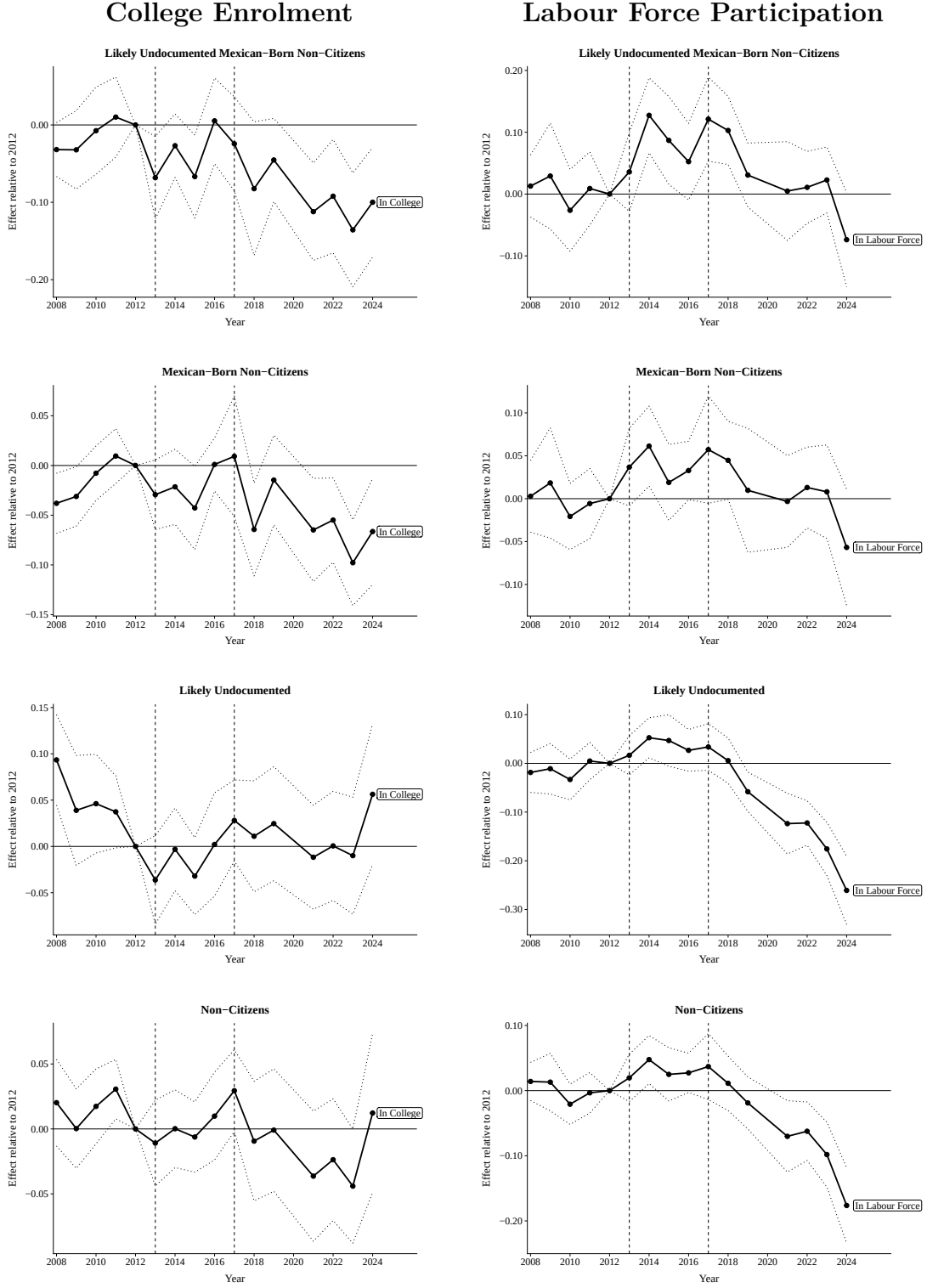
Notes: Entries are linear probability model coefficients for labour force participation, reported in percentage points. The sample consists of likely undocumented Mexican-born non-citizens in the American Community Survey. “DACA-Eligible” identifies individuals satisfying the observable DACA eligibility criteria; because DACA receipt is not observed, estimates are intent-to-treat effects of eligibility. The row “DACA-Eligible” reports the baseline pre-DACA difference between likely eligible and ineligible individuals. The interaction rows report period-specific changes for likely DACA-eligible individuals relative to likely ineligible individuals and relative to the pre-DACA baseline. Models add fixed effects, state-specific linear trends, and demographic controls cumulatively as indicated. Standard errors are in parentheses and are clustered at the state level. $p < 0.1$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

Table 8: Event Study Estimates for College Enrolment and Labour Force Participation

	<i>In College</i>	<i>In Labour Force</i>
DACA-Eligible $\times$ 2008	-0.0319 (0.0178)	0.0129 (0.0256)
DACA-Eligible $\times$ 2009	-0.0322 (0.0258)	0.0292 (0.0437)
DACA-Eligible $\times$ 2010	-0.0074 (0.0288)	-0.0263 (0.0338)
DACA-Eligible $\times$ 2011	0.0102 (0.0263)	0.0090 (0.0303)
DACA-Eligible $\times$ 2013	-0.0682* (0.0272)	0.0358 (0.0325)
DACA-Eligible $\times$ 2014	-0.0268 (0.0209)	0.1274*** (0.0309)
DACA-Eligible $\times$ 2015	-0.0667* (0.0273)	0.0868* (0.0362)
DACA-Eligible $\times$ 2016	0.0053 (0.0283)	0.0524 (0.0316)
DACA-Eligible $\times$ 2017	-0.0243 (0.0309)	0.1212*** (0.0347)
DACA-Eligible $\times$ 2018	-0.0824 (0.0440)	0.1029*** (0.0281)
DACA-Eligible $\times$ 2019	-0.0453 (0.0274)	0.0307 (0.0264)
DACA-Eligible $\times$ 2021	-0.1121** (0.0321)	0.0049 (0.0408)
DACA-Eligible $\times$ 2022	-0.0921* (0.0374)	0.0108 (0.0297)
DACA-Eligible $\times$ 2023	-0.1356*** (0.0375)	0.0228 (0.0272)
DACA-Eligible $\times$ 2024	-0.0999** (0.0361)	-0.0739 (0.0388)
N	32,199	32,199

Notes: Entries report coefficients from DACA event-study specifications for likely undocumented Mexican-born non-citizens in the American Community Survey. Coefficients are reported in probability units, so 0.01 corresponds to 1 percentage point. Each row gives the difference between likely DACA-eligible and likely ineligible individuals in that calendar year relative to the omitted reference year, 2012. Because DACA receipt is not observed, estimates should be interpreted as intent-to-treat effects of DACA eligibility. The year 2020 is excluded. All specifications include demographic controls, state fixed effects, year fixed effects, and state-specific linear trends. Standard errors are clustered at the state level. $\cdot p < 0.1$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

Figure 2: Event-study estimates by sample group

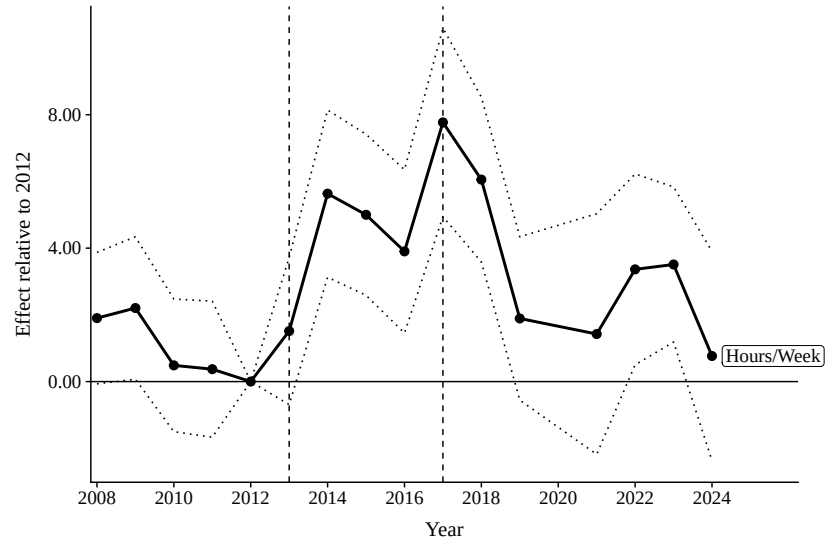


Notes: Each plot reports DACA event-study estimates using a different population, restricted to likely undocumented Mexican-born non-citizens, Mexican-born non-citizens, likely undocumented immigrants, and non-citizens. All individuals are on the education margin defined in Section 4. Coefficients are reported in probability units and are plotted relative to the omitted reference year, 2012. Dotted lines report pointwise 95 percent confidence intervals. Vertical dashed lines mark the beginning of DACA implementation and the beginning of the uncertainty period.

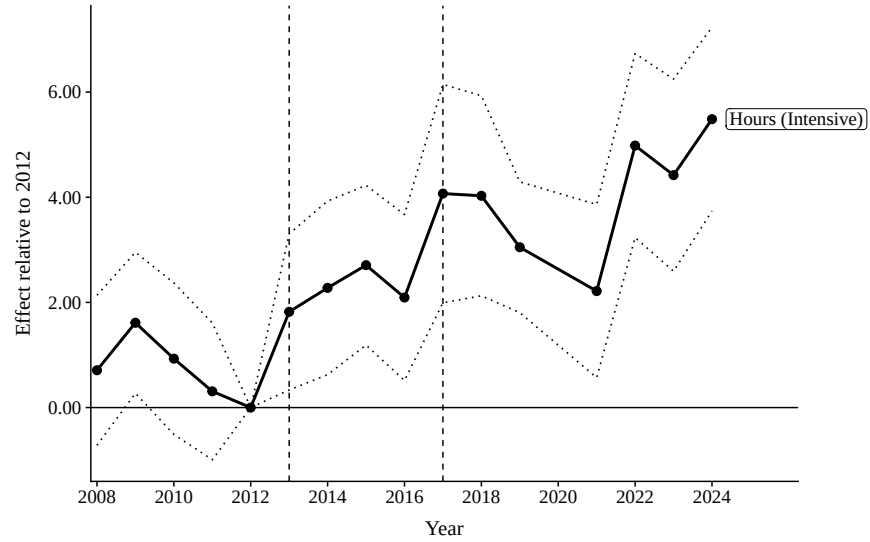
Table 9: Hours Worked Responses to DACA

	<i>Outcome</i>	
	Hours/Week (Full Sample)	Hours/Week (Intensive)
DACA-Eligible	-1.3337*** (0.3548)	-1.0798*** (0.3013)
DACA-Eligible $\times$ Implementation	2.9371*** (0.6434)	1.4648** (0.4469)
DACA-Eligible $\times$ Uncertainty	3.0543*** (0.5949)	3.1482*** (0.5473)
Age (+ Age Squared)	Y	Y
Sex	Y	Y
Marital Status	Y	Y
Race	Y	Y
Year of Immigration	Y	Y
State FE	Y	Y
Year FE	Y	Y
State-specific trends	Y	Y
Clustered SE	State	State
N	32,199	23,289

Notes: Entries are linear regression coefficients for weekly hours worked. “Hours/Week (Full Sample)” uses the full analytic sample, with non-workers assigned zero hours. “Hours/Week (Intensive)” restricts the sample to individuals with positive hours worked. The sample consists of likely undocumented Mexican-born non-citizens in the American Community Survey. “DACA-Eligible” identifies individuals satisfying the observable DACA eligibility criteria; because DACA receipt is not observed, estimates are intent-to-treat effects of eligibility. The row “DACA-Eligible” reports the baseline pre-DACA difference between likely eligible and ineligible individuals. The interaction rows report period-specific changes for likely DACA-eligible individuals relative to likely ineligible individuals and relative to the pre-DACA baseline. All specifications include demographic controls, state fixed effects, year fixed effects, and state-specific linear trends. Standard errors are clustered at the state level. $\cdot p < 0.1$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.



(a) Hours Worked per Week



(b) Hours Worked per Week Conditional on Employment

Figure 3: Event study estimates for hours worked per week, with 2012 as the reference year.

Notes: Points plot coefficients from DACA event-study specifications for likely undocumented Mexican-born non-citizens in the American Community Survey. Panel (a) uses weekly hours worked in the full analytic sample, with non-workers assigned zero hours. Panel (b) restricts the outcome to workers with positive hours. Coefficients are measured in weekly hours and are plotted relative to the omitted reference year, 2012. Dotted lines report pointwise 95 percent confidence intervals. Vertical dashed lines mark the beginning of DACA implementation and the beginning of the uncertainty period. Specifications include demographic controls, state fixed effects, year fixed effects, and state-specific linear trends, with standard errors clustered at the state level.

Table 10: Pre-treatment Event-Study Estimates for College Enrolment and Labour Force Participation Responses to In-State Resident Tuition for Undocumented Immigrants

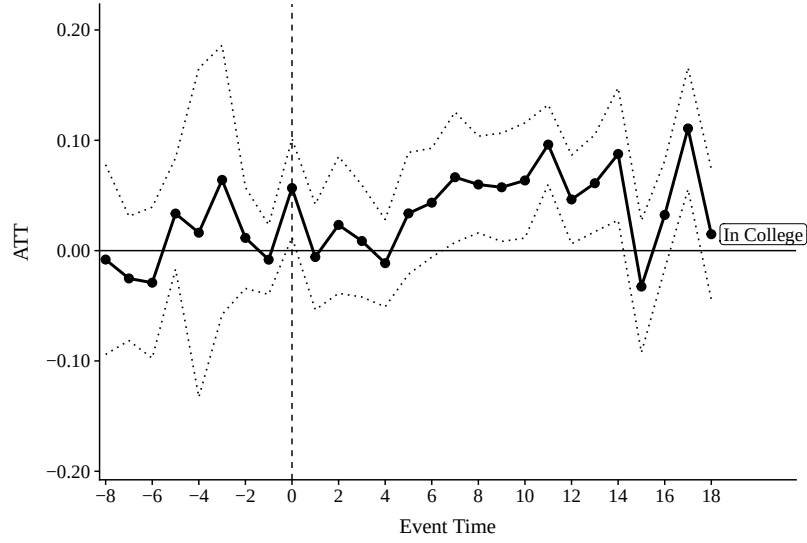
<i>Event Time</i>	<i>In College</i>	<i>In Labour Force</i>	<i>Event Time</i>	<i>In College</i>	<i>In Labour Force</i>
-22	0.0192 (0.0148)	0.0390 (0.0292)	-11	-0.0398 (0.0644)	0.1259 (0.0849)
-21	-0.0423 (0.0221)	0.1391* (0.0592)	-10	0.0168 (0.0454)	-0.0478 (0.0810)
-20	0.0844 (0.0510)	-0.1965 (0.1014)	-9	-0.0411 (0.0637)	0.0120 (0.0872)
-19	-0.1289*** (0.0210)	0.1259 (0.0863)	-8	-0.0081 (0.0439)	-0.0211 (0.0376)
-18	-0.0012 (0.0529)	0.0962 (0.0810)	-7	-0.0252 (0.0288)	-0.0041 (0.0308)
-17	0.1080*** (0.0240)	-0.0062 (0.0335)	-6	-0.0291 (0.0349)	0.0506* (0.0208)
-16	0.0139 (0.0269)	-0.0833** (0.0271)	-5	0.0336 (0.0257)	0.0145 (0.0240)
-15	-0.1106* (0.0553)	0.0823 (0.0755)	-4	0.0162 (0.0761)	-0.0268 (0.0484)
-14	-0.0471 (0.0295)	-0.0350 (0.0533)	-3	0.0641 (0.0622)	-0.0052 (0.0520)
-13	0.0677 (0.0416)	-0.0816 (0.0418)	-2	0.0116 (0.0234)	0.0459 (0.0475)
-12	0.0236 (0.0566)	-0.0056 (0.0484)	-1	-0.0082 (0.0161)	-0.0100 (0.0236)
<i>Pre-treatment Average</i>	-0.0010 (0.0046)	0.0095 (0.0109)			

Notes: Entries report pre-treatment event-study average treatment effects on the treated (ATTs) from the Callaway and Sant'Anna staggered-adoption difference-in-differences estimator for repeated cross-sections. The sample consists of likely undocumented Mexican-born non-citizens in the American Community Survey. Event time is measured relative to the first year in which a respondent's state offers in-state resident tuition to undocumented immigrants; negative event times are years before treatment. Coefficients are reported in probability units, so 0.01 corresponds to 1 percentage point. Comparison units are individuals in never-treated or not-yet-treated states. The final row reports the average pre-treatment ATT across all listed leads. Standard errors are reported in parentheses and clustered at the state level. $\cdot p < 0.1$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

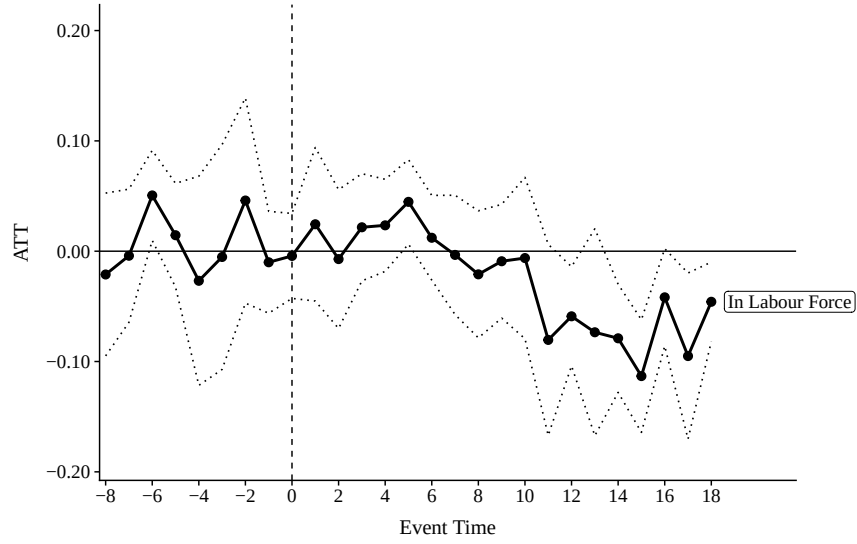
Table 11: Post-treatment Event-Study Estimates for College Enrolment and Labour Force Participation Responses to In-State Resident Tuition for Undocumented Immigrants

<i>Event Time</i>	<i>In College</i>	<i>In Labour Force</i>	<i>Event Time</i>	<i>In College</i>	<i>In Labour Force</i>
0	0.0567* (0.0228)	-0.0043 (0.0196)	10	0.0636* (0.0266)	-0.0062 (0.0373)
1	-0.0058 (0.0244)	0.0244 (0.0355)	11	0.0961*** (0.0184)	-0.0804 (0.0442)
2	0.0233 (0.0317)	-0.0072 (0.0321)	12	0.0464* (0.0205)	-0.0590* (0.0229)
3	0.0087 (0.0259)	0.0217 (0.0248)	13	0.0611** (0.0224)	-0.0735 (0.0479)
4	-0.0114 (0.0202)	0.0234 (0.0213)	14	0.0876** (0.0306)	-0.0790** (0.0250)
5	0.0338 (0.0282)	0.0447* (0.0195)	15	-0.0326 (0.0305)	-0.1132*** (0.0260)
6	0.0435 (0.0252)	0.0122 (0.0196)	16	0.0323 (0.0250)	-0.0418 (0.0225)
7	0.0665* (0.0301)	-0.0034 (0.0276)	17	0.1107*** (0.0281)	-0.0951* (0.0384)
8	0.0599** (0.0224)	-0.0210 (0.0294)	18	0.0149 (0.0302)	-0.0459* (0.0184)
9	0.0574* (0.0251)	-0.0091 (0.0263)			
<i>Post-treatment Average</i>	0.0428** (0.0143)	-0.0270 (0.0154)			

Notes: Entries report post-treatment event-study average treatment effects on the treated (ATTs) from the Callaway and Sant'Anna staggered-adoption difference-in-differences estimator for repeated cross-sections. The sample consists of likely undocumented Mexican-born non-citizens in the American Community Survey. Event time is measured relative to the first year in which a respondent's state offers in-state resident tuition to undocumented immigrants; event time 0 is the first treatment year. Coefficients are reported in probability units, so 0.01 corresponds to 1 percentage point. Comparison units are individuals in never-treated or not-yet-treated states. The final row reports the average post-treatment ATT across all listed lags. Standard errors are reported in parentheses and clustered at the state level. $\cdot p < 0.1$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.



(a) College Enrolment



(b) Labour Force Participation

Figure 4: Event-study estimates for responses to in-state resident tuition for undocumented immigrants. Panel (a) reports college enrolment, and panel (b) reports labour force participation.

Notes: Points plot event-study average treatment effects on the treated (ATTs) from the Callaway and Sant'Anna staggered-adoption difference-in-differences estimator for repeated cross-sections. The sample consists of likely undocumented Mexican-born non-citizens in the American Community Survey. Treatment is defined as living in a state-year in which undocumented students are eligible for in-state resident tuition. Event time is measured relative to the first treatment year; event time 0 is the first year of in-state resident tuition access, negative values are pre-treatment leads, and positive values are post-treatment lags. Coefficients are reported in probability units. Dotted lines report pointwise 95 percent confidence intervals. Comparison units are individuals in never-treated or not-yet-treated states.

Table 12: College Enrolment and Labour Force Participation Responses to In-State Resident Tuition for Undocumented Immigrants by Calendar Year

<i>Year</i>	<i>In College</i>	<i>In Labour Force</i>	<i>Year</i>	<i>In College</i>	<i>In Labour Force</i>
2001	0.0605*** (0.0099)	-0.0018 (0.0115)	2011	0.0560* (0.0248)	-0.0219 (0.0283)
2002	0.0601* (0.0273)	-0.0078 (0.0321)	2012	0.0558 (0.0350)	-0.0049 (0.0438)
2003	-0.0188 (0.0278)	0.0460 (0.0405)	2013	0.0952*** (0.0188)	-0.1034** (0.0400)
2004	0.0218 (0.0349)	-0.0070 (0.0350)	2014	0.0380 (0.0293)	-0.0285 (0.0387)
2005	-0.0096 (0.0283)	0.0114 (0.0303)	2015	0.0563** (0.0201)	-0.0738 (0.0377)
2006	-0.0031 (0.0217)	0.0347 (0.0197)	2016	0.0816 (0.0438)	-0.1047*** (0.0280)
2007	0.0432 (0.0304)	0.0436 (0.0231)	2017	-0.0167 (0.0413)	-0.1119*** (0.0331)
2008	0.0327 (0.0300)	0.0177 (0.0197)	2018	0.0637 (0.0436)	-0.0291 (0.0286)
2009	0.0748* (0.0315)	-0.0001 (0.0313)	2019	0.0830* (0.0400)	-0.1094*** (0.0318)
2010	0.0602* (0.0244)	-0.0126 (0.0282)			
N	68,987		N	68,987	

Notes: Entries report calendar-time average treatment effects on the treated (ATTs) from the Callaway and Sant'Anna staggered-adoption difference-in-differences estimator for repeated cross-sections. The sample consists of likely undocumented Mexican-born non-citizens in the American Community Survey. Each row averages treatment effects across treated cohorts observed in the listed calendar year. Coefficients are reported in probability units, so 0.01 corresponds to 1 percentage point. Comparison units are individuals in never-treated or not-yet-treated states. Standard errors are reported in parentheses and clustered at the state level. $p < 0.1$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

Table 13: College Enrolment and Labour Force Participation Responses to In-State Resident Tuition for Undocumented Immigrants by DACA Policy Regime (Including Post-COVID)

	<i>In College</i>	<i>In Labour Force</i>
ATT (Pre-DACA)	3.6152 (1.9031)	0.8115 (1.7401)
ATT (Implementation/Stability)	6.7794*** (1.7835)	-7.7614** (2.6913)
ATT (Uncertainty, Pre-COVID)	4.3361* (2.0448)	-8.3468*** (2.0645)
ATT (Uncertainty, Post-COVID)	-22.3194*** (3.0751)	12.3759*** (2.5176)
Age (+ Age Squared)	Y	Y
Sex	Y	Y
Marital Status	Y	Y
Race	Y	Y
Year of Immigration	Y	Y
N	74791	74791

Notes: Entries report calendar-time aggregated average treatment effects on the treated (ATTs) from the Callaway and Sant'Anna staggered-adoption difference-in-differences estimator for repeated cross-sections, reported in percentage points. The sample consists of likely undocumented Mexican-born non-citizens in the American Community Survey. Treated individuals are those living in states that offer in-state resident tuition to undocumented immigrants; comparison units are individuals in never-treated or not-yet-treated states. The pre-DACA period covers 2001–2012, the implementation/stability period covers 2013–2016, the pre-COVID uncertainty period covers 2017–2019, and the post-COVID uncertainty period covers 2021–2024. The year 2020 is excluded. Post-COVID estimates are included as a robustness exercise and are not the preferred specification. Standard errors are clustered at the state level. $p < 0.1$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

Table 14: College Enrolment and Labour Force Participation Responses to In-State Resident Tuition by DACA Policy Regime by Treatment Timing Definition

	<i>In College</i>			<i>In Labour Force</i>		
	(1)	(2)	(3)	(1)	(2)	(3)
ATT (Pre-DACA)	3.6152 (1.9031)	3.6152 (1.9031)	4.2130* (2.1168)	0.8115 (1.7401)	0.8115 (1.7401)	1.3648 (1.9079)
ATT (Implementation)	6.7794*** (1.7835)	6.9249*** (1.9077)	7.1068*** (1.9344)	-7.7614** (2.6913)	-4.6924* (2.0469)	-4.4728* (2.1538)
ATT (Uncertainty)	-10.8956*** (2.2312)	-11.1023*** (2.3086)	-10.9680*** (2.3879)	3.4948 (1.8102)	7.5935*** (2.2825)	7.0839** (2.3849)
Age (+ Age Squared)	Y	Y	Y	Y	Y	Y
Sex	Y	Y	Y	Y	Y	Y
Marital Status	Y	Y	Y	Y	Y	Y
Race	Y	Y	Y	Y	Y	Y
Year of Immigration	Y	Y	Y	Y	Y	Y
N	74791	74379	73543	74791	74379	73543

Notes: Entries report calendar-time aggregated average treatment effects on the treated (ATTs) from the Callaway and Sant’Anna staggered-adoption difference-in-differences estimator for repeated cross-sections, reported in percentage points. The sample consists of likely undocumented Mexican-born non-citizens in the American Community Survey. “(1)” column defines treatment using the first year in which a state offers in-state resident tuition to undocumented immigrants. “(2)” column defines treatment using the first year in which a state offers in-state resident tuition to DACA recipients. “(3)” column drops Arkansas, Massachusetts, Ohio, and Virginia, where treatment timing differs across definitions or is otherwise sensitive to coding. The uncertainty-period average excludes 2020. Standard errors are clustered at the state level. $p < 0.1$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

Table 15: College Enrolment and Labour Force Participation Responses to In-State Resident Tuition for Undocumented Immigrants by Calendar Year (Including Post-COVID)

<i>Year</i>	<i>In College</i>	<i>In Labour Force</i>	<i>Year</i>	<i>In College</i>	<i>In Labour Force</i>
2001	0.0605*** (0.0099)	-0.0018 (0.0115)	2013	0.0952*** (0.0188)	-0.1034** (0.0400)
2002	0.0601* (0.0273)	-0.0078 (0.0321)	2014	0.0380 (0.0293)	-0.0285 (0.0387)
2003	-0.0188 (0.0278)	0.0460 (0.0405)	2015	0.0563** (0.0201)	-0.0738 (0.0377)
2004	0.0218 (0.0349)	-0.0070 (0.0350)	2016	0.0816 (0.0438)	-0.1047*** (0.0280)
2005	-0.0096 (0.0283)	0.0114 (0.0303)	2017	-0.0167 (0.0413)	-0.1119*** (0.0331)
2006	-0.0031 (0.0217)	0.0347 (0.0197)	2018	0.0637 (0.0436)	-0.0291 (0.0286)
2007	0.0432 (0.0304)	0.0436 (0.0231)	2019	0.0830* (0.0400)	-0.1094*** (0.0318)
2008	0.0327 (0.0300)	0.0177 (0.0197)	2021	-0.2291*** (0.0330)	-0.1468** (0.0503)
2009	0.0748* (0.0315)	-0.0001 (0.0313)	2022	-0.1641*** (0.0365)	0.1509** (0.0552)
2010	0.0602* (0.0244)	-0.0126 (0.0282)	2023	-0.5695*** (0.0482)	0.6288*** (0.0621)
2011	0.0560* (0.0248)	-0.0219 (0.0283)	2024	0.0699 (0.0556)	-0.1378* (0.0605)
2012	0.0558 (0.0350)	-0.0049 (0.0438)			
N	74,791		N	74,791	

Notes: Entries report calendar-time average treatment effects on the treated (ATTs) from the Callaway and Sant’Anna staggered-adoption difference-in-differences estimator for repeated cross-sections. The sample consists of likely undocumented Mexican-born non-citizens in the American Community Survey. Each row averages treatment effects across treated cohorts observed in the listed calendar year. Coefficients are reported in probability units, so 0.01 corresponds to 1 percentage point. Comparison units are individuals in never-treated or not-yet-treated states. The table includes post-COVID years 2021–2024 and excludes 2020. Standard errors are reported in parentheses and clustered at the state level. $p < 0.1$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.